Linnæus University
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Master of Science Thesis

Detection of NORM contamination in LNG systems

A Systems Engineering literature review



Author: Nazrin Babashova
Supervisor: Per Lindström
Examiner: Michael Strand
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Linnæus University
Sweden

Abstract

Naturally occurring radioactive material (NORM), which is present in the oil and gas industry, can pose a risk to the workers engaged in the activities in this field. Nevertheless, to our knowledge, there is no study on NORM presence and NORM measurement in liquefied natural gas (LNG) fuel systems and installations. Therefore, the question on whether NORM, particularly radon and radon progeny, can pose a risk to the health of employees, is still open. This study provides the initial base for further investigation and real NORM measurement. In this study, all stages of radon and radon decay products movement and deposition starting from natural gas production till the final destination are analyzed and reviewed. Several techniques for NORM detection and measurement are proposed. In the case if radon levels exceed the reference level, several actions towards radon reduction are proposed.

Key words

LNG, TENORM, NORM, Detection, Contamination, Radon, Systems Engineering.

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Nomenclature

Abbreviations

AAV	Ambient Air Vaporizer
ACD	Activated Charcoal Detector
AGRU	Acid Gas Removal Unit
BOG	Boil-off Gas
CNG	Compressed Natural Gas
EIC	Electret ion chamber
EPA	Environmental Protection Agency
GDU	Gas Dehydration Unit
GSU	Gas Sweetening Unit
GTL	Gas to Liquids
GTP	Gas to Power
GTS	Gas to Solids
GWP	Global Warming Potential
HHV	Higher Heating Value
HP	High Pressure
IAEA	International Atomic Energy Agency
IOGP	International Association of Oil & Gas Producers
LNG	Liquefied Natural Gas
LPG	Liquefied Petroleum Gas
LSA	Low Specific Activity
LWD	Logging While Drilling
MCHE	Main Cryogenic Heat Exchanger
MRU	Mercury Removal Unit
MWD	Measurement While Drilling
NG	Natural Gas
NGL	Natural Gas Liquid
NIOSH	National Institute for Occupational Safety and Health
NORM	Naturally Occurring Radioactive Material
PPE	Personal Protective Equipment
PTS	Port-to-ship bunkering method

SRU	Sulfur Recovery Unit
STS	Shore-to-ship bunkering method
TENORM	Technologically Enhanced Naturally Occurring Radioactive Materials
TGTU	Tail Gas Treating Unit
TL	Thermoluminescence
TLD	Thermoluminescent Dosimeter
TTS	Truck-to-ship bunkering method

Units

Bq	Becquerel: the rate of radioactive decay equal to one disintegration per second
Bq/(Nm^3)	Becquerel per normal meter cube: the unit of measurement for the radioactivity within a volume of space
Bq/g	Becquerel per gram: specific activity/activity per quantity of a radionuclide
Gy [J/kg]	Gray: the unit of absorbed dose/absorbed energy per unit mass of matter
K	Kelvin: a base unit of temperature
Pa	Pascal: the unit for pressure
Sv [J/kg]	Sievert: the unit of equivalent and effective radiation dose
Sv/h	Sievert per hour: dose rate/dose per unit time
atm	The standard atmosphere: standard pressure/Earth's atmospheric pressure at sea level
kg/m^3	Kilogram per meter cube: a standard unit of material density
m^3	Meter cube: the unit of volume
Nm^3	Normal meter cube: the value that a matter of a constant mass occupies under normal or standard conditions
ppmv	Part per million by volume/a unit of concentration
tonne	The unit of mass equal to 1000 kg

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1 Introduction

1.1 Background and problem description

Liquefied Natural Gas (LNG) fuel, which is an oil and gas industry product, has captured particular attention recently. The reason behind increased attention to LNG fuel systems is current environmental issues resulting in demand for less greenhouse effect polluting fuels in the maritime and land-based industries. Compared to other fuels, LNG-based combustion processes produce fewer sulfur emissions; therefore, LNG is considered as a possible fuel option [1]. However, according to the life-cycle study of different marine fuels [2], where global warming potential (GWP) of fuels over 20-year and 100-year time frame was evaluated, it was found that LNG positive impact on climate is either questionable or LNG releases even more emissions than other types of marine fuel. Nevertheless, over the past two decades, LNG technologies have been significantly developed resulting in an increase of LNG technologies efficiency [3]. Therefore, LNG is considered as a fuel of the future, and there is a huge potential of LNG technologies to be improved further. Various ferry and shipping companies mentioned by [4] such as Destination Gotland, Furetank, and Viking Line have already adopted LNG-fueled engines. Besides, a new LNG tank terminal is now projected in the port of Oxelösund, Sweden [5]. However, as it was mentioned before, LNG is the oil and gas industry product which implies the possibility of naturally occurring radioactive material (NORM) contamination of LNG fuel systems. NORM is the term referred to material containing radionuclides which can be found in the environment. The radionuclides of interest are uranium-238 (^{238}U), uranium-235 (^{235}U) and thorium-232 (^{232}Th), and their radioactive decay products such as radium and radon. NORM can present in various industries such as the coal industry and metal mining, including the oil and gas industry [6]. LNG fuel systems onboard vessels may contain radon and radon decay products which may have an adverse effect on human health. There is no short-term effect caused by NORM presenting in LNG fuel systems; nevertheless, exposure to radiation increases the risk of long-term effects such as cancer [7].

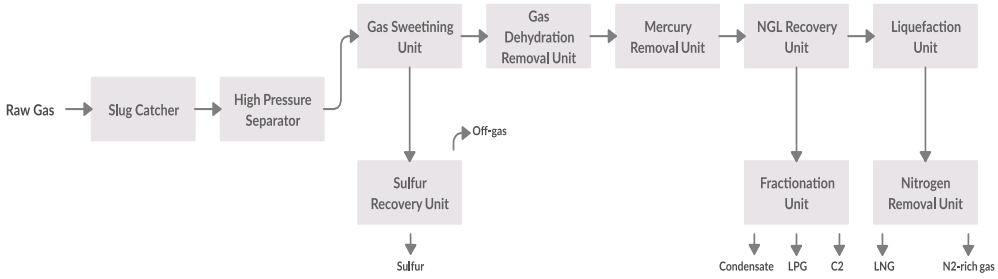


Figure 1 Typical LNG refinery diagram [51].

The reason behind the presence of NORM in final LNG product, particularly radon and radon decay products, is due to radon presence in sedimentary rocks from where

oil and gas is usually extracted. After extraction, numerous treatment steps take place, that can be seen in Figure 1, such as absorption via amine solvents in order to remove particular gases such as carbon dioxide and hydrogen sulfide, elimination of water vapor with the help of dehydration process where molecular sieve is used, and removal of mercury commonly via charcoal columns [8]. During treatment steps, radon and radon decay products, particularly Pb-210 [9], can be accumulated on molecular sieve or any other used material. Radon and radon progeny, particularly Po-210 and Pb-210 [8], can also be captured by charcoal columns. In addition, some amount of radon tends to accumulate in liquefied petroleum gas (LPG) [10], which is a derivative of natural gas as it can be seen in Figure 1. Nevertheless, despite purification processes, certain concentrations of radon decay products in LNG were recorded [11]. The range of radon concentration varied a lot starting from less than 1 pCi/L and reaching up to 23 pCi/L. Therefore, there is a possibility of radon and radon progeny presence in LNG. Moreover, taking into consideration the accumulative nature of radon decay products, namely the accumulation of radon daughters on the inner parts of LNG equipment, there is a risk of health-threatening levels of radiation to which workers can be exposed.

According to the Swedish law, the employer shall examine the working conditions and evaluate possible risks of any person being affected by ill-health or accidents at work on a regular basis [12]; and is responsible for implementing the measures necessary to prevent ill-health and accidents at work [13]. Taking into consideration all the aforementioned facts, the importance of possessing information regarding NORM presence in LNG fuel systems is apparent. However, according to [4], *"the Swedish government has little, if any, knowledge about whether NORM is present in LNG fuel systems"*. This statement is further supported by the e-mail correspondence between the Swedish national agencies: Linnaeus University, Swedish Radiation Safety Authority, Swedish Work Environment Authority, Swedish Transport Agency, and Swedish Energy Agency [14] – [31]. Therefore, a thorough investigation is required in order to address gaps in knowledge regarding the existence of NORM in LNG fuel systems and provide a safe work environment.

1.2 Hypothesis

Taking into consideration the aforementioned facts, the author postulates the following hypothesis:

There is a risk of NORM presence and contamination of LNG systems, and there are certain locations in LNG systems where radon and radon progeny are more likely to be observed. Therefore, there is a risk of radiation levels posing a threat to a safe work environment.

1.3 Research questions

The main research questions of this study are:

I. How can NORM contamination be detected in LNG fuel systems?

II. Where are radon and radon decay products more likely to be accumulated?

To be able to answer the main research questions, the following sub-research questions should be reviewed:

III. What kind of techniques can be used in order to identify NORM contamination level in LNG fuel systems?

IV. What kind of standards and procedures should be followed in order to obtain reliable results?

V. How do radon and radon progeny particles tend to behave in LNG systems?

1.4 Limitations

The study examines NORM contamination and NORM measurement procedures in LNG fuel system, i.e. LNG tank, LNG terminal, pipelines, pumps, and other associated equipment. The study is limited to the following aspects and areas:

I. The study does not address NORM measurement procedures of other systems where NORM can be present such as LNG refinery.

II. The study does not consider NORM measurement procedures of LNG engine emissions, i.e. the study is focused on NORM contamination of LNG equipment.

1.5 Research methodology

This study is based on a literature review methodology. The literature review is focused on the scientific engineering aspects that have been gathered in regard to the topic, concept, or phenomena. With the help of the literature review, it is possible to link already existing works between each other and establish the relationship between different theories. In addition, the literature review may give a rise to a new hypothesis which should be investigated in more depth [32].

There are four stages of literature review, namely [32]:

1. The formulation of the problem, i.e. what kind of question is being raised and what is the matter with the raised question.
2. Literature search itself, i.e. the thorough research of the materials related to the problem.
3. The evaluation of the found information, i.e. which material or materials play the most significant role in understanding the problem, or which material or materials add up the value to the study.

4. Analysis and interpretation, i.e. the discussion and overall conclusion of the gathered materials.

In this study, all four stages are followed:

1. The problem of radiation due to NORM accumulation in LNG systems is being raised. The matter with NORM deposition is that it may pose a threat to the health and endanger a safe work environment.
2. Excessive literature search on the topics of NORM, radiation, LNG, and NORM measurement has been conducted.
3. All gathered works and papers were evaluated, and the most useful ones were used in this paper.
4. Different papers were analyzed, for instance, the papers covering the behavior of particles of different sizes in Newtonian fluid, and, based on these papers, some conclusions were drawn regarding radon and radon progeny particles.

Apart from the theoretical literature review, the philosophy of the Design for Six Sigma (DFSS) process of the scientific engineering method 'Systems Engineering' was applied to this study [33] – [35]. First and foremost, according to the concept of DFSS, the question of whether process or product exists should be raised. Apart from questions that were already asked, the question of whether any established process on NORM measurement in LNG exists was raised in this study. Since, to our knowledge, there has been no study carried on NORM deposition measurement in LNG systems, i.e. the answer to the aforementioned question is "no", it is important to identify the process according to DFSS concept. Besides, according to DFSS philosophy, the first stage is identification, where numerous stages are included such as review and assessment of applicable technologies. In this study, different techniques were reviewed and analyzed. The second stage, namely design, is about the selection of the best concepts and preparation of preliminary designs that are aligned with the Voice of the Customer (VoC). The Voice of the Customer (VoC) is a research method utilized by businesses to describe the preferences of the customers. In this study, not only several technologies were reviewed, but also compared between each other, and the best concepts were chosen.

1.5.1 Philosophy of Science

Apart from the 'Systems Engineering' approach, it is also important to look at the matter from the scientific philosophy point of view. For instance, in a recently presented B.Sc. thesis [36], apart from the purely engineering approach, the underlying philosophy of science also was taken into consideration.

One of the philosophical movements is called logical positivism [37]. Logical positivism is an approach that is based on facts and pure logic. In order to identify whether a statement is truthful or not, verifiability principle, or verificationism, is used. According to the verifiability principle, a statement has meaning in the case if it is possible to verify the statement empirically. Despite the fact that logical positivism has faced criticism and been rejected by modern philosophers since the

verification principle cannot be verified itself, it still had a great impact, especially on social science [38].

Another philosophical movement is pragmatism. Pragmatism, which is also called pragmaticism, is the philosophical movement where the importance of practical consequences is emphasized. In simple terms, something can be considered as true if it works. One of the founders of pragmatism, Charles Sanders Peirce, wrote in his work [39]: *"Consider the practical effects of the objects of your conception. Then, your conception of those effects is the whole of your conception of the object."*

Peirce also distinguished between different methods of data collection, namely deduction, induction, and abduction. With the help of deductive logic, which is also called a "top-down" approach, all collected data is narrowed down to a specific conclusion. In the case of inductive logic, a number of observations is used in order to "catch" a pattern to establish a general statement which may or may not be true. The third logic, namely abduction logic, was developed by Peirce. Abduction reasoning helps to come to the best possible conclusion out of many alternative conclusions to explain emerged facts and incomplete observations; however, the conclusion which is derived via abductive logic is not necessarily true one [40].

This study is based on the philosophy of the logical positivism since:

1. This study established a hypothesis that is based on the pure logic.
2. The hypothesis, or statement, should be verified empirically, i.e. with the help of the experiment.

This study is based on the deductive logic since:

1. Lots of theories are gathered and put together.
2. Based on the gathered theories, a new hypothesis is established.
3. In order to address the hypothesis, observations should be collected.
4. In the case if collected observations approve the initial hypothesis, then the theory can be confirmed. Otherwise, in the case if observations contradict the theory, then the theory is rejected.

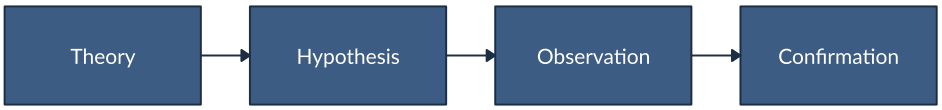


Figure 2 Deductive reasoning steps [40].

In this study, the first two steps of deductive reasoning, which is illustrated in Figure 2, are utilized. For the remaining two steps, NORM level measurement in LNG systems and derived experimental data are required.

2 Theory

2.1 NORM, radon and radiation types

2.1.1 NORM and TENORM

According to the International Atomic Energy Agency (IAEA) Safety Glossary [41], naturally occurring radioactive material is defined as *"a radioactive material containing no significant amounts of radionuclides other than naturally occurring radionuclides"*. NORM, among which are uranium-238 (^{238}U), uranium-235 (^{235}U) and thorium-232 (^{232}Th) which have been mentioned before, have been present in the Earth's crust since the Earth formation. Radionuclides are not stable and tend to undergo radioactive decay process after which nuclides become stable and stop emitting radiation. Nevertheless, due to the fact that some radioactive elements decay slowly, i.e. these elements have a long half-life, they are still present in rock and soil. Due to certain human activities such as oil and gas extraction, naturally occurring radioactive materials tend to concentrate or, in other words, tend to become "technologically enhanced" and are exposed to the human environment. This phenomenon is known as Technologically Enhanced Naturally Occurring Radioactive Material (TENORM), and the National Academy of Sciences defines TENORM as *"any naturally occurring radioactive materials not subject to regulation under the Atomic Energy Act whose radionuclide concentrations or potential for human exposure have been increased above levels encountered in the natural state by human activities"*. According to [42], total amounts of produced TENORM in the United States per year may exceed 1 billion; thus, TENORM constitutes a challenge for particular industries.

2.1.2 Radon and radiation types

There are several different isotopes of radon; however, the one which poses a threat to human health is radon-222. Radon-222 (^{222}Rn), which is a result of decay processes chain starting from uranium-238 (^{238}U), is a radionuclide with a half-life of approximately 3.8 days [6]. The decay chain can be seen in Figure 3.

The decay rate or activity of any radioactive element can be measured in becquerels (Bq), where 1 Bq is one radioactive disintegration per second, or in curie (Ci), or to be more precise picocuries (pCi) - 2.2 disintegrations per minute. Thus, in order to measure the activity concentration of radon, the unit of Bq/m^3 or pCi/L is used [43]. Radon is a colorless inert gas which is soluble in water and organic solvents. Despite the fact that radon is gas, its decay products are solid particles that can be inhaled and deposited in the lung increasing the potential for lung cancer. Radon daughters being accumulated on the lung emit a different type of radiation, namely alpha and beta radiations as well as weak gamma radiation [6].

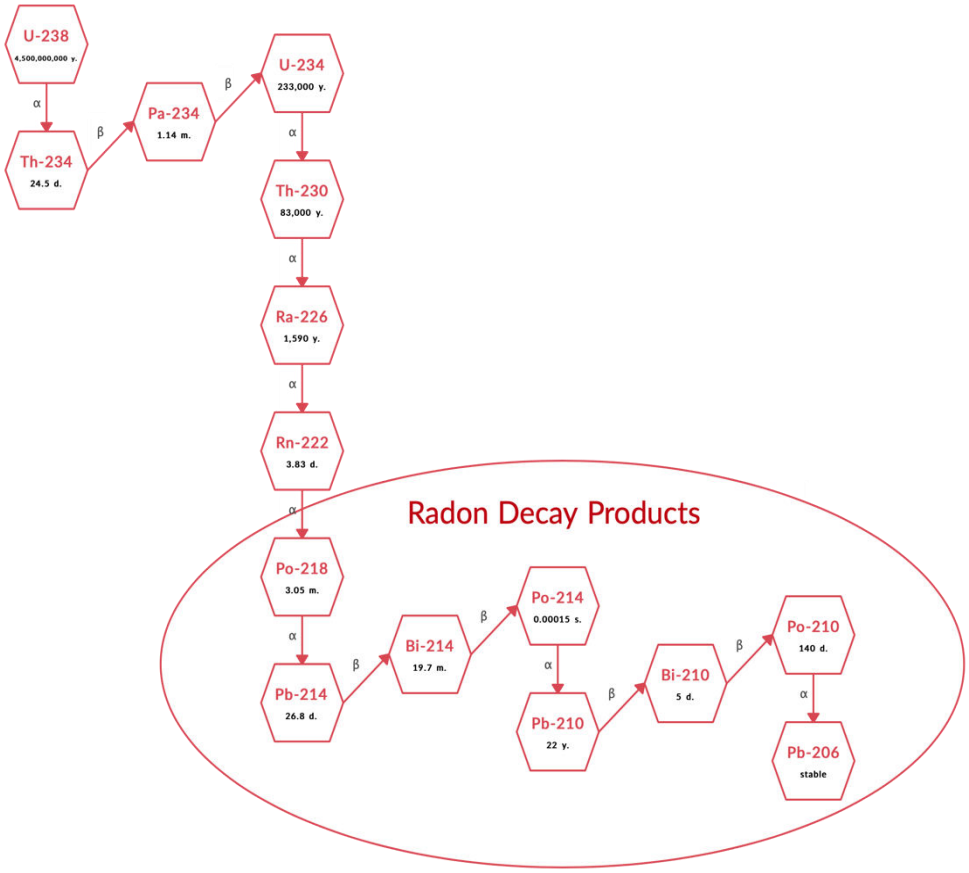


Figure 3 Uranium-238 decay chain [4].

Alpha (α), beta (β), and gamma (γ) decay are based on three fundamental forces, namely the strong force, the weak force and the electromagnetic force. In all three cases, the decay increases the stability of the nucleus due to released protons and neutrons. During alpha decay, an alpha particle which is a helium nucleus with two protons and two neutrons, is emitted [44].

$$ {}^{A}_{Z}X = {}^{A-4}_{Z-2}X' + {}^{4}_{2}He \qquad (1) $$

Where:
X – parent nucleus
X' – daughter nucleus
A – atomic mass
Z – atomic number
He – helium

Beta decay can be divided into two different forms: beta plus decay (β^+) and beta minus decay (β^-). During beta plus decay, proton inside a radionuclide nucleus is converted into a neutron, and neutrino together with positron which is an electron with positive electric charge are emitted. Beta plus decay occurs in radionuclides

rich in protons, and this type of emission decreases proton number leading to the balance between protons and neutrons [44].

$$ {}^{A}_{Z}X = {}_{Z-1}{}^{A}X' + e^{+} + v_e \qquad (2) $$

Where:
e^+ – positron
v_e – neutrino

In the case of beta minus decay, the number of neutrons outweighs the number of protons. Neutrons are converted into a proton, and electron together with antineutrino are released in the process [44].

$$ {}^{A}_{Z}X = {}_{Z+1}{}^{A}X' + e^{-} + \overline{v_e} \qquad (3) $$

Where:
e^- – electron
$\overline{v_e}$ – antineutrino

During alpha and beta decay, nuclei are in excited states. The excited states of nuclei tend to decay into lower states, and this process is accompanied by electromagnetic radiation emission in the form of gamma rays [44].

$$ {}^{A}_{Z}X = {}^{A}_{Z}X' + {}^{0}_{0}\gamma \qquad (4) $$

Different types of radiation have different penetrating power. Alpha particles can travel only a few centimeters and do not constitute a danger since alpha radiation cannot penetrate the skin. However, alpha radiation can be harmful when radioactive material emitting alpha particles is inhaled, swallowed, or absorbed through open wounds. Beta particles can be stopped by a piece of clothing. However, compared to alpha particles, beta particles can penetrate human skin to some extent and cause skin injuries if the skin is exposed to beta radiation for a long period of time. Gamma radiation can travel large distances and has the strongest penetration power [45]. Among all aforementioned types of radiation, alpha and beta radiation are the ones of interest since alpha and beta particles can be inhaled during the maintenance process.

2.2 Natural gas exploration and treatment

2.2.1 Natural gas exploration process

In order to find out whether natural gas reserves are present in rock, geologists use several methods. First of all, there is an observation of surface rocks called geological examination or geological survey giving an idea about underground formations. In the mid 1800s, it was understood that certain slopes called anticlinal slopes where the earth has folded up on itself have a greater likelihood of containing natural gas compared to other areas. Natural gas can move toward the surface of the earth thus a large portion of natural gas, more specifically methane, will dissipate

into the air. However, a significant amount of natural gas will also be trapped under impermeable formation of rock through which natural gas will not be able to escape. There are various types of such formations; however, the most common one is dome shape impermeable rock layers which look like an umbrella capturing the natural gas trying to reach the surface, and which is illustrated in Figure 4. Therefore, a geological survey is an essential step which helps to explore possible natural gas deposits. Once a location that has a probability to contain natural gas or petroleum reserves is set, further tests can be carried in order to obtain more data regarding area with natural gas potential [46].

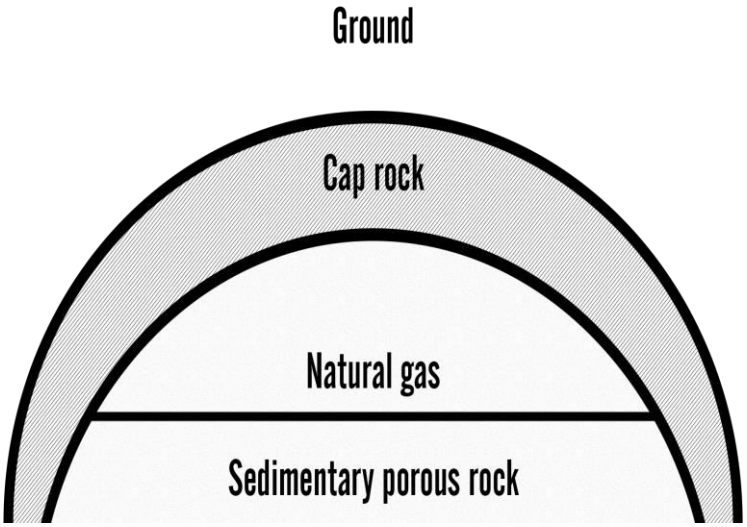


Figure 4 Rock formation [46].

Seismic survey is one of the further steps which should be taken to get more detailed data. In old days, explosive materials were used such as dynamite in order to set small explosions and create seismic waves which further were caught by a special device called geophones, and received data was interpreted by geophysicists, geologists, and petroleum engineers. Nevertheless, nowadays, due to technology improvement, non-explosive seismic technology is used which also helps to avoid any environmental damage. Special vehicle is carrying equipment which creates vibrations, and, in turn, these vibrations create seismic waves which help to receive useful data. The same approach is applied for offshore exploration where instead of the truck there is a ship, and instead of the geophone, there is hydrophone [46].

In addition to the seismic survey, there are other means of obtaining data such as the use of magnetometers. The earth's magnetic field is not uniform and differs depending on location. The difference in magnetism can be caused by several reasons one of which is the differing nature of rock formations. Therefore, oil and gas deposits may affect the magnetic field, and variance can be observed via magnetometer – a device measuring even the small variance in the magnetic field [46].

When obtained data shows a high probability of petroleum and gas reserves in the formation rocks, exploratory well can be drilled in order to get more precise data on

deposits. Afterward, the logging procedure, which is an ongoing record of the physical characteristics of rock and fluid with the help of the logging tool, is carried.

There are over a hundred types of logging, and some of the most common are standard logging and electric logging. During standard logging, the physical properties of a well can be evaluated. For instance, the drill cuttings which are pieces of rock that are generated when a well is drilled, allow geologists to analyze the material. In the case of electric logging, electric current is run through the rock layers measuring the electrical resistance. The electrical logging helps geologists to obtain data regarding fluid content and properties. Recently, a novel method which is logging while drilling (LWD), also called measurement while drilling (MWD), provides the same data about the well as previously mentioned techniques. However, instead of lowering the logging tool into the well at the end of wireline cable, the logging tool is incorporated into the drill string [46]. In the case if logging provides positive data regarding oil and natural gas deposits, production process takes place.

2.2.2 Production of natural gas and radon escape mechanism

Oil and gas can be found in any porous rock but in most of the cases it is discovered in sedimentary rocks such as sandstone or limestone. Usually, the oil and gas reservoirs are located on the top of a layer of interstitial water. Uranium, which is a radioactive element, is present in the sedimentary rock [6]. One of the reasons behind the presence of uranium, to be more precise uranyl ion, is the physical adsorption of uranyl ion on the surface of organic compounds which are also present in sedimentary rock [47]. Two of uranium decay products, namely radium and radon, can escape from the rock and be dissolved in the interstitial fluids. One of the main reasons behind radon emanation, i.e. release of radon from the mineral grain to the pore space where radon can diffuse further, is based on recoil phenomenon. During radium decay, alpha particle, which is helium nucleus, is ejected. When the alpha particle is ejected, radon atom which has been formed recoils in the opposite direction which can be observed in Figure 5 [48].

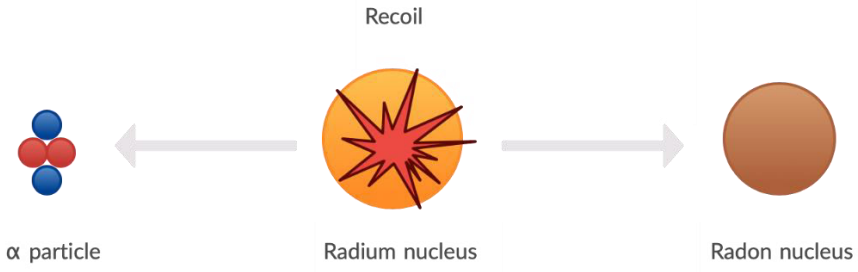


Figure 5 Radium nuclide decay mechanism [48].

The process resembles a recoil of the rifle after the shot. The release of radon to the pore space depends on several factors: the location of radium atom, i.e. whether it is close or far from the surface of a mineral sample, and direction of recoiled radon atom, i.e. whether it recoils in direction to the surface of the grain or against the surface. When all conditions are favorable, i.e. radium atom is located near the

surface of the mineral grain, and radon recoils toward the surface, radon atom can escape to the pore. However, frequently, due to the fact that recoiled radon atom has substantial energy, it can travel all the way through the pore space, and enter in nearby mineral grains. Nevertheless, in the case if pore space is filled with water, radon tends to remain in pore space. Due to the higher density compared to air, water slows down the speed of radon thus preventing radon atom to travel further and eventually captures it [48]. The process is illustrated in Figure 6.

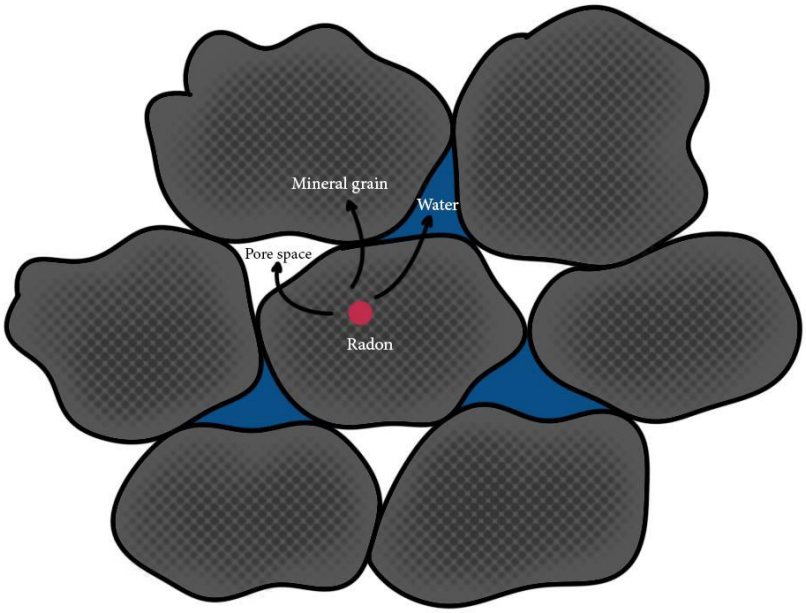


Figure 6 Radon atom possible movements [48].

It is also worth mentioning that radon is gas, in contrast to uranium and radium which are solids, implying that radon more easily escapes mineral grain and then moves further with natural gas in pore space. In order to retrieve the gas trapped in rock formations, well should be drilled, and certain equipment is installed in the well such as well casing, wellhead, and installation of lifting equipment. The equipment plays a crucial role for further steps: for instance, well casing contributes to the strengthening of the wellbore sides, and natural gas can be brought to the surface without being lost by entering to rock layers on the way up. After all the necessary procedures are completed, well may start to produce natural gas. Natural gas located in pressurized formations will naturally rise up through established well, especially in the case when a deposit consists of only natural gas or together with a light condensate. This case is called a primary recovery. However, sometimes special treatment is needed in order to provide an effective flow of hydrocarbons through the well. Such treatment may include injection of water, gases such as CO_2, or an acid such as hydrochloric, into the well to open up the formation and allow hydrocarbons to flow through [46]. One of the most common technologies is hydraulic fracturing: the pressure of injected fluid, which is usually the mixture of water (99.5%), sand and chemical additives (0.5%) “cracks” the fractures which are already present in the formation ensuring the release of trapped natural gas. Apart from water, as it has been mentioned before, carbon dioxide can also be used to crack the formation which has even more advantages compared to water as injected

fluid. Injected CO_2 and sand mixture fractures the formation through which desired hydrocarbons can flow through. Afterward, CO_2 is vaporized, and only sand remains in the formation, holding the cracks, which have been enlarged and created by fracturing, open. Such a technique does not generate underground wastes, and also ensures protection of groundwater resources [49]. Another common technique is horizontal drilling: with the help of horizontal drilling, the area of a well can be increased without creating several vertical wells. One of the reasons behind the use of horizontal drilling is that the significant part of oil and gas deposits is concentrated in a horizontal dimension rather than in a vertical dimension. Therefore, with the help of horizontal drilling more natural gas can be extracted [50]. The extraction of natural gas is known as the production process or upstream. After extraction, natural gas transportation, or midstream, takes place. In the case of LNG, natural gas is first converted to LNG in the production plant which is near to the extraction area, and afterward is transported to the market, and all stages will be deeply examined in the following chapters.

2.2.3 Natural gas treatment

After extraction, raw natural gas cannot be transported directly and should undergo some treatment processes such as removal of water vapor and solids, as well as natural gas liquids such as ethane, propane, and butane. After treatment, there are several ways to transport natural gas: pipelines, liquefied natural gas (LNG), compressed natural gas (CNG), gas to solids (GTS), gas to power (GTP), gas to liquids (GTL), and other ways [46]. Nevertheless, among all the aforementioned methods, only LNG and pipelines are fully commercial and mature technologies. GTL is commercial but requires certain improvements in order to enhance efficiency, and CNG together with GTS have not been put on a commercial scale yet. Therefore, natural gas is usually transported either by pipelines or by LNG cargo ships or trucks. The main criteria of choosing a particular method are based on distance and volume of extracted natural gas. LNG becomes a reasonable solution when natural gas should be transported for long distances, particularly for the routes which should be crossed via ships since undersea pipeline construction is not cost-efficient. In addition, in order to transfer a large amount of natural gas for long distances via pipelines, large diameter pipes and gas recompression facilities are needed which is also costly [51]. Nowadays, LNG is a fast-growing market, and, according to BP Statistical Review of World Energy 2019, the amount of natural gas transferred via LNG cargo ships and trucks in 2018 was 431 billion cubic meters whereas in 2008 it was only 234.9 billion cubic meters, which shows the substantial increase of LNG trade. The data is show in Table 1. Compared to 2017, LNG supplies increased by almost 10% (37 bcm) in 2018 due to the fact that new liquefaction plants in Australia, the United States and Russia were either started or the number of plants increased [52].

Table 1 Pipeline and LNG inter-regional trade of natural gas in billion cubic meters [52].

Worldwide (bcm)	**2008**	**2009**	**2010**	**2011**	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**
Inter-regional pipeline trade	527.5	420.5	435.6	466.9	460.3	459.5	436.1	445.6	479.6	510.5	512.4
LNG trade*	234.9	249.7	302.4	328.3	324.9	326.8	333.6	337.1	358.3	393.9	431.0

*LNG exports include re-exports.

The traditional supply chain can be observed in Figure 7. First, as it has already been mentioned, natural gas is extracted and then supplied to LNG refinery. Afterward, LNG is transported via cargo ships or trucks to the demand point, where LNG can be used further for numerous purposes such as using LNG as marine fuel or regasification of LNG and using both as gas or electricity [51]. In Figure 7, LNG chain where LNG is used for electricity production is illustrated.

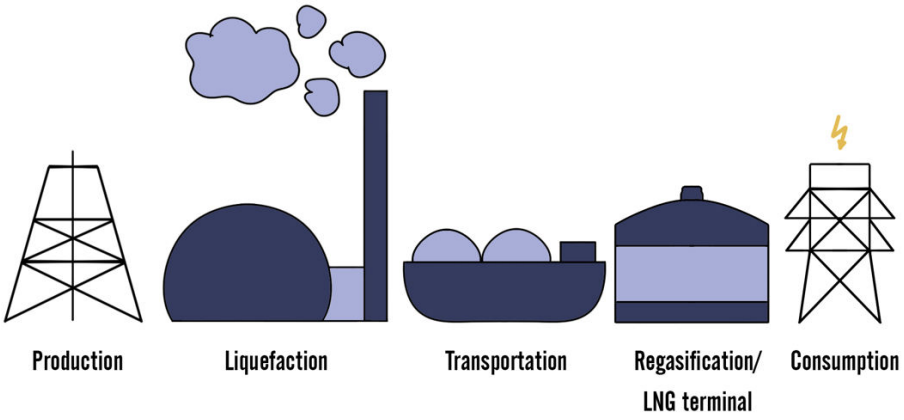


Figure 7 The main elements of a typical LNG supply chain [51].

The traditional LNG plant is illustrated in Figure 1. Typically, first, the liquids from natural gas should be removed; therefore, the raw natural gas is sent to a slug catcher. Afterward, gas is sent to a high pressure (HP) separator whereas the liquids are fed to a medium pressure separator, where the hydrocarbon liquids are further separated and fed to the condensate stabilizer. The liquids which were fed to the condensate stabilizer are separated, and formed condensate comprises of C_5 and other heavy hydrocarbons. As it can be seen, vapors from the medium separator and stabilizer column overhead are compressed and recycled back to the HP separator.

The vapors then are fed to the Gas Sweetening Unit, GSU which is also called Acid Gas Removal Unit (AGRU), where hydrogen sulfide (H_2S) and carbon dioxide (CO_2) are removed. The hydrogen sulfide is removed with the help of an amine solvent in order to meet sulfur product requirement which is typically below 4 ppmv [51]. It is important to remove H_2S since it contributes to general corrosion, which is also called uniform corrosion; pitting corrosion, which causes the formation of cavities in the material; and acid corrosion, which results in crack or blisters. In addition, H_2S can be emitted through crack having a detrimental impact on people's health and damaging to the environment [53]. Another compound, which is CO_2, should be decreased to 50 ppmv in order to avoid CO_2 freezing which will cause heat exchanger plugging and other operational problems [54].

The acid gas is fed to the Sulfur Recovery Unit (SRU) which is usually comprised of a Claus unit and a Tail Gas Treating Unit (TGTU), and the off-gas from the TGTU is sent to the atmosphere. The sweet gas from GSU is sent to the Gas

Dehydration Unit (GDU) in order to decrease the amount of water vapor to below 0.1 ppmv [51]. Water should be removed since water vapor freezes and may damage the main liquefaction exchanger [55]. For GDU process, molecular sieves are utilized. In hot climate operation, the water content in the sweet gas can be substantial; therefore, the sweet gas can be chilled first in order to remove most of the water prior to the molecular sieve unit. This procedure, i.e. the sweet gas chilling, is more energy-efficient and cost-effective rather than direct feeding of saturated sweet gas to the molecular sieve unit. As it can be seen from the scheme, there is also Mercaptans Removal Unit where mercaptans from the dried gas can be removed via molecular sieve in order to meet the sulfur standards [51].

Afterward, the gas is sent to the Mercury Removal Unit (MRU), where mercury (Hg) must be reduced to less than 10 nanograms per cubic meter. The presence of mercury has a detrimental effect on brazed-aluminum heat exchangers that are used in cryogenic systems causing equipment corrosion used. On top of that, mercury can be a threat to the environment and safety. There are numerous methods of mercury removal, and one of the most common methods is the use of the sulfur which reacts with mercury and form a stable compound on the sorbent surface. The support material is usually activated carbon, and sulfur-impregnated activated carbon is an established commercial process in natural gas refineries [51]. Nevertheless, other trapping agents such as metal sulfides, silver and gold, and other support materials or carrier agents such as alumina and zeolite can be used [56].

After MRU, treated gas is fed to Natural Gas Liquid (NGL) Recovery Unit where C_2+ or C_3+ are removed and recovered, and a lean gas, i.e. a gas which contains mainly methane, is further liquefied. The use of NGL Recovery Unit excludes the need in the scrubber column where aromatics and heavy hydrocarbons are removed in order to avoid waxing phenomena in the main exchanger. There can be also ethane recovery where ethane is recovered and further supplied to the petrochemical refinery. The heavy hydrocarbons, namely C_3-C_5+ liquids, are also separated: propane and butane can be exported as separate products, or as liquefied petroleum gas (LPG), and C_5+ and heavier components can be used for gasoline blending [51].

After NGL Recovery Unit, methane gas is fed to the Liquefaction Unit where a refrigeration process occurs. During liquefaction process, a refrigerant takes the heat from the gas: the process is based on the principle of a refrigeration cycle where a refrigerant undergoes cyclical expansion and compression, and turns the gas into liquid by decreasing its temperature to cryogenic one [51]. The lighter hydrocarbons, or lean gas, is fed to the main cryogenic heat exchanger (MCHE) in order to convert lean gas to LNG. Nevertheless, not only lean gas but undesired heavy compounds may also enter MCHE which may constitute a substantial operational problem. These heavy compounds can freeze and accumulate in the MCHE, leading to the decrease of the heat transfer efficiency and increase of operational costs. In the case of the continued build-up of solids in the MCHE, a blockage and a shutdown of the plant may occur [57]. In addition, there is a case study on MCHE failure which was caused by mercury embrittlement leading to the shutdown of refrigeration compressors for 6 weeks which, in turn, led to the loss of LNG production [58].

After the lean gas is liquefied, a nitrogen removal is required in the case if the nitrogen content exceeds the commercial specification of LNG which is typically below 1 mole per cent. There are several reasons behind the need for nitrogen removal: the presence of nitrogen in LNG does not allow to meet a higher heating value (HHV) requirement set for LNG product; and may cause so-called "rollover" phenomenon. Nitrogen has no positive impact on HHV and needs to be removed so LNG product specification can be met. In addition, due to the fact that natural gas trade is based on a heat content rather than on weight, nitrogen fills up valuable space, and thus should be eliminated. Another problem that appears when nitrogen content is higher than 1 mole per cent is called a rollover. The floor and walls of the tank are heated, and the heat is absorbed by the tank is transmitted, or more specifically, convected to the liquid. At the top layer, light gases such as nitrogen start to evaporate, and the liquid in the tank becomes denser. The evaporation taking place in LNG tanks is known as boil-off gas (BOG). When boil-off occurs, convection takes place where a bottom layer of warm and less dense liquid goes up to the interface. However, evaporation does not occur since the top layer exerts hydrostatic pressure on the bottom layer. Since top layer, or lighter layer, gets denser, and bottom layer, or heavier layer, gets lighter, at some point the density of two layers approach each other, and rollover occurs [51]. The process is illustrated in Figure 8. According to [59], even a little difference in temperature (less or around 1 K) causes a sudden revolution of density layers in the tank. Due to the fact that the bottom layer is superheated, after rollover a substantial amount of vapor is released. One of the major risks of rollover is the over-pressurization of the tank. There is also a possibility of the failure of tank relief system due to the high evaporation rate which will eventually lead to LNG discharge into the atmosphere [51].

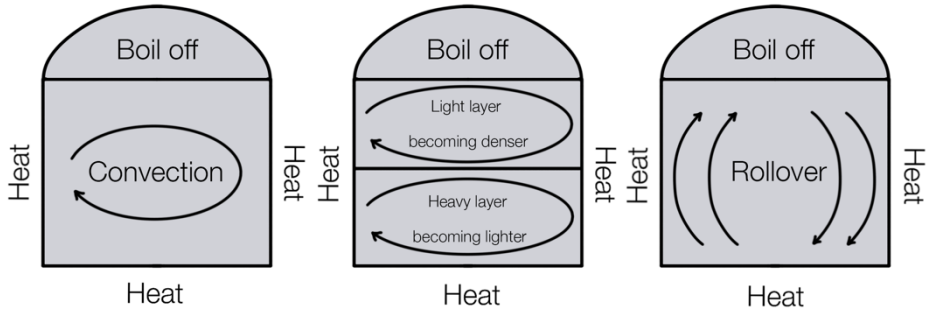


Figure 8 The stages of rollover phenomenon in LNG tank [59].

After all treatment processes, LNG is loaded either to trucks or to ships which are equipped with special cryogenic tanks and transported to the demand point.

2.3 Radon presence in LNG refinery and LNG product

2.3.1 Radon in LNG refinery

According to the review study [60], where records on radon and radon decay products were gathered, the concentration of radon in natural gas varies between 5 Bq/m^3 and 200,000 Bq/m^3. Prior to entering refinery, according to [8], a certain

amount of radon decay products, which were formed during transportation via pipeline since it takes time for natural gas to reach LNG plant, accumulates either on inner part of pipeline or at plant inlet filters. Radon, due to the fact that it is in gaseous state, is able to path through filter and enter to LNG refinery [8]. During certain treatment processes, radon and radon progeny can also be removed from natural gas. For instance, Pb-210 which is radon daughter, can be captured via molecular sieve during Gas Dehydration Unit process. According to the study [9], where the concentration of Pb-210 deposition on molecular sieve was estimated taking into consideration the adsorption equilibrium coefficients (0.037 for molecular sieve), the concentration of radon in the natural gas (100 to 20,000 Bq/Nm^3), the typical operating conditions in LNG refinery (5.5 MPaA) and the operation period of the adsorbents from its installation to exchange, the following data was obtained. In this research, two different standards were taken into account: IAEA NORM regulation with 1.0 Bq/g exemption level, and the International Association of Oil & Gas Producers (IOGP) NORM regulation with 0.2 Bq/g exemption level. The amount of accumulated Pb-210 has a positive correlation both with radon concentration in natural gas and the number of sieve operating years. In the case if radon concentration equals to 400 Bq/Nm^3, even after 10 years of operation the level of deposited Pb-210 will not exceed 0.2 Bq/g. Nevertheless, in the case of 2,000 Bq/Nm^3 Rn-222 concentration, the level of Pb-210 accumulation is below 1.0 Bq/g after 10 years of operation; however, at this radon concentration, it exceeds 0.2 Bq/g after 3 years. In case if radon concentration is 10,000 Bq/Nm^3, it is over 1.0 Bq/g after 3 years. Experiment was also conducted on silica gel, and in the case of Rn-222 concentration of 10,000 Bq/Nm^3, the level of Pb-210 accumulation exceeds 1.0 Bq/g after one year [9]. Conducted experiment illustrates that sufficient amount of radon progeny can be found in treatment units.

Another treatment unit where radon daughters tend to concentrate is Mercury Removal Unit. Radon progeny can be accumulated on activated charcoal bed, particularly Po-210 and Pb-210, due to the fact that the chemical properties of Po-210 and Pb-210 are similar to those of mercury. Radon itself will also be trapped by charcoal [8]. Radon can also be captured by zeolite [61], one of the materials which is used in MRU. Nevertheless, compared to many types of materials such as molecular sieve, silica gel and alumina, charcoal possesses greater adsorption properties and capable of capturing greater amount of radon [62].

Afterward, remained radon and radon decay products move to NGL unit. According to numerous data from natural gas refineries, severe NORM contamination of inner parts of pumps was observed in this unit, particularly in propane/propene fraction. Other types of equipment were affected as well, and thin invisible films of Pb-210 were detected and reported [60]. Apart from invisible Pb-210 films, dark grey or black films can be observed in natural gas refineries due to contamination of short-lived decay products of radon, namely Po-218, Pb-214, Bi-214 and Po-214, on inner parts of equipment. Some of these short-lived daughters are the reason of high-energy gamma radiation emission and, consequently, the reason of external gamma radiation dose rates in the area of contaminated equipment [63]. In addition, low specific activity (LSA) sludge, which is a term for radioactive deposit, can be observed in different places such as pipeline bottom; nevertheless, LSA sludge tends to deposit in voluminous equipment such as NGL storage tanks, separator and

dehydration vessels, LPG delivery facilities. The deposition occurs mostly due to the long retention time [60].

After NGL unit, as it has been mentioned in the previous chapter, the heavy hydrocarbons such as propane and butane are separated making up liquefied petroleum gas (LPG) which is a derivative of natural gas. Radon tends to accumulate in LPG due to the fact that the boiling point of radon, which is shown in Table 2, is between that of ethane and propane [10]. The radon concentration of derived LPG varies from 70 to 48,000 Bq/m^3 [64]. Due to LPG derivation, 30 to 75% of radon can be eliminated from natural gas [10].

Table 2 Different boiling points of LPG components [10].

LPG component	**Boiling point (°C)**
Methane	-161.5
Ethane	-88.3
Radon	-61.8
Propane	-42.2
Butane	-0.5

After LPG has been derived, the rest amount of radon and radon progeny enters Liquefaction Unit. As it has been noted before, rich-methane gas enters to MCHE where it is converted to LNG, and where unwanted components can be accumulated. To our knowledge, there is no study on radon and radon decay products deposition in cryogenic units. Nevertheless, it can be assumed that a certain amount of radon enters MCHE and also is able to deposit in MCHE. The reasons behind assumptions are:

I. **Presence of radon in the final LNG product.** There is a record of radon decay products presence in LNG [11] which means that a certain amount of radon enters MCHE and other cryogenic units and ends up in final product.

II. **Accumulation of radon in MCHE due to its increased density.** To our knowledge, there is no information regarding the density of radon at LNG temperatures, i.e. -169°C [51]. Nevertheless, there is a data on radon density in the gaseous state at standard conditions (0°C, 1 atm) which is 9.73 kg/m^3, and density at normal boiling point (-62°C) which is around 4400 kg/m^3 [65]. Thus, the density of radon tends to increase with temperature decrease (Table 3), and the physical state of radon changes during LNG production from gas to liquid, and finally to solid since the freezing point of radon is -71°C [66]. Due to the increased density of radon particles, the settling velocity, which is also known as fall velocity or terminal velocity, increases [68], which facilitates the settling process of radon particles.

Table 3 Densities of saturated liquid radon [66].

T (K)	300	290	280	270	260	250	240	230	220	211	205	202
ρ (kg/m^3)	3557	3650	3744	3824	3906	3990	4076	4163	4250	4330	4382	4408

The case of shutdown in natural gas refineries should also be taken into consideration. After the shutdown of a natural gas plant, the continuous flow of natural gas stops; therefore, short-lived radon daughters decay since their half-life varies between 160 microseconds and 27 minutes [8]. Thus, there is no gamma radiation emission; however, the long-lived decay products of radon such as Pb-210, Bi-210 and Po-210, remain on the inner part of the equipment. These radionuclides pose no threat due to weak gamma radiation but the alpha particles emitted by Po-210 and Pb-210 pose a potential hazard if they are ingested or inhaled during the maintenance process [63]. Experiment conducted by [67] showed that during maintenance of LPG equipment an employee can be exposed to a significant amount of radiation. It was observed that a disassembled pump impeller from LPG refinery emitted a significant amount of combined gamma and beta radiation, and even after a year of storage impeller emitted a considerable number of alpha particles. The concentration of 30 pCi/L was detected during twelve minutes of maintenance whereas the limit for constant occupational exposure to Pb-210/Po-210 is 0.2 pCi/L for 40 hours per week.

It is also worth mentioning that Pb-210 concentration was discovered in LPG rail tanks [69]. This study indicates the fact that NORM contamination can be deposited not only on natural gas facilities but also further on transportation equipment. Taking into consideration all aforementioned facts such as radon concentration in natural gas feed which can be substantially high, radon decay products deposition in treatment units and transportation facilities, it is apparent that radon concentration level in LNG should be investigated.

2.3.2 Final LNG product and radon concentration

LNG mostly consists of methane which is between 87 mole per cent to 99 mole per cent. Other compounds such as ethane, propane, butane, as well as nitrogen and trace amounts of sulfur (less than 4 ppmv), and CO_2 (around 50 ppmv) are also present in LNG. The content of LNG depends on the content of the initial gas source and the treatment processes to which gas was subjected. In general, LNG is an odorless, colorless, nontoxic and also noncorrosive cryogenic liquid (-169°C) at 1 atm pressure. LNG produces very low particle emissions, i.e. emissions contain almost no sulfur oxides and low level of nitrogen oxides. In addition, compared to other hydrocarbon fuels, LNG releases substantially lower carbon emissions. Despite the fact that LNG is non-toxic, it may lead to asphyxiation in the case of an unventilated area where there is an absence of oxygen. In addition, LNG can be ignited in the presence of the right concentrations of air [51].

To our knowledge, there has been only one record of radon and radon decay products presence in LNG [11]. According to the conducted study, the range of radon concentration varied a lot starting from less than 1 pCi/L and reaching up to 23 pCi/L. The concentration of radon and radon decay products in LNG is based on the following assumptions:

I. **Radon concentration in the initial feed.** It is assumed that there is a direct relation between initial radon concentration and radon concentration in the final LNG product.

II. **Natural gas flow rate in LNG plant.** In the case of turbulent flow, some of lead atoms may stick to the inner surface of transportation system [70]. Therefore, it is assumed that the type and speed of flow in natural gas facility is directly related to the NORM concentration in final LNG product.

III. **The type of material used in treatment units.** Due to the fact that different materials have different adsorption power, the concentration of radon in the final LNG product depends on how much of radon can be adsorbed during treatment processes.

It is also assumed that even with insignificant radon and radon progeny concentration in LNG, over time radon and radon decay products concentration in LNG systems will increase due to the depositary nature of radon and radon daughters. Taking into consideration all facts which have been mentioned, it is extremely important to possess information regarding radon and radon decay products concentration deposited on the inner parts of equipment to ensure proper provision of safe work environment. In the following chapters, LNG systems such as LNG carrier, as well as maintenance procedure, together with information regarding where radon is more likely to be accumulated, will be examined in depth.

2.4 LNG systems and maintenance

2.4.1 LNG carrier and NORM contamination

The capacity of LNG ships may range between less than 30,000 m^3 to around 265,000 m^3. However, the common capacity varies between 125,000 m^3 and 140,000 m^3 which can contain from 58,000 to 65,000 tonnes of LNG. In order to transport LNG, special tank design is required which keeps LNG at cryogenic temperature and pressure which is just above atmospheric pressure. LNG ships are used in the case when a consumer is located in the coastal area. In the case of the inland location, LNG road tankers with double-skinned, vacuum insulated tanks are used [51].

To our knowledge, there is no study conducted on NORM measurement and no records on NORM contamination of LNG carriers and tanks. Nevertheless, there is a study on deposition of Pb-210 in LPG rail cars, and author noted that *"each batch of LPG will add new daughter activity to the dust except for that which remains in the gas itself and is discharged from the tank when it reaches its destination"*, highlighting the depository nature of radon daughters [69]. It is assumed that radon and radon decay products tend to deposit more in LNG land-based storage tanks rather than in LNG ship tanks. The assumptions are based on the following facts:

I. **Radon density.** LNG storage temperatures are much lower (-169°C) compared to LPG temperatures. LPG storage temperatures and pressures vary; nevertheless, LPG temperature does not go below -43°C [71]. At

LPG temperatures, radon is either in the gaseous or the liquid state whereas in LNG radon is in the solid state; thus, the density of radon is much higher in LNG compared to LPG. Therefore, radon and radon decay products are more easily settled in LNG tank.

II. **Sea motion**. It is assumed that radon and radon decay products tend to deposit more in LNG land-based storage tanks rather than in LNG ship tank. The reason behind the assumption is a motion of cargo during transportation, to be more precise the motion of sea. The motion of sea definitely affects the processes going on in LNG tank since the phenomenon of rollover which has been mentioned before does not constitute a problem for LNG cargo due to mixture of layers by sea motion [51]. Therefore, it can be assumed that it is more difficult for radon and radon decay solid particles to settle in LNG carrier tanks at sea, and part of them remain "floating" in LNG.

III. **Truck motion**. The same assumption regarding the mixture of layers in LNG can be applied to LNG truck. During straight road and constant movement, it is assumed that LNG does not experience any shake and stays relatively stable. Nevertheless, when truck stops, starts movement or takes a turn, due to inertia phenomenon, LNG experiences certain movement. Therefore, particles tend to float in LNG; nevertheless, some portion of radon and radon decay daughters deposit on equipment both in LNG cargo and LNG truck.

Despite the fact that rollover is not considered as a problem for LNG carrier, there is a case study on Moss Type LNG carrier which experienced the rollover phenomena. Nevertheless, there were several reasons behind occurred BOG such as loading of tanks with heavy LNG whereas a heel inside was of lighter density [72]. Under normal conditions, rollover does not pose an issue both for LNG cargo and LNG road trailer [51].

When LNG cargo/LNG truck is unloaded, as it has been mentioned by [69], some amount of radon and radon daughters is discharged as well. One of the reasons behind is high flow rate where methane and other particles "kick out" deposited radon and radon progeny, and radon together with radon decay products follow LNG stream and appear in LNG storage tanks or directly in LNG fueled ship depending on whether it is shore-to-ship (STS)/port-to-ship (PTS) or truck-to-ship (TTS) bunkering method.

2.4.2 LNG systems contamination

In order to identify the locations with a high probability of radon and radon decay products, it is important to possess information on the properties of radon and radon progeny and LNG. The behavior of solid particles in fluid depends on numerous parameters such as solid properties, such as size and density; on the properties of the fluid, such as density and viscosity; on the operation parameters of pipe flow such as flow velocity, flow direction and diameter of pipe [73]. According to study [74], where authors measured the size of radon decay particles in different environments and different modes, namely unattached and attached, diameter of radon and radon

daughter particles did not exceed 1 µm. Therefore, radon and radon decay products are considered as submicron particles or nanoparticles.

There are various factors that affect particle deposition such as gravitation, electrostatic interactions, gravitational forces, turbulence, and Brownian diffusion. However, Brownian diffusion is more significant for nanoparticles than for larger particles, leading to higher deposition rates. Thermophoresis is also more important for nanoparticles compared to other factors, and the smallest particles are affected the most due to their low inertia [75]. Gravitational forces are not dominant for nanoparticles: as it has been mentioned before, Brownian and diffusional forces together with thermophoresis have more effect on the deposition rate of nanoparticles. However, there is a study on radon and radon progeny deposition [76] where it is stated that the deposition rate of radon and radon progeny particles which are smaller than 100 nm was dominated by diffusion. For particles larger than 500 nm, the main force that affected sedimentation rate was a gravitational force. There is another study where it is stated that in the case if particles have high density, gravity can become a significant driving force even if the size of particles is less than 100 nm [77]. Taking into consideration the fact that the density of radon and radon progeny particles is significantly high, and the diameter of particles varies greatly starting from 1nm in the case if radon and radon decay products are unattached, it is extremely challenging to predict the exact behavior of radon and radon daughters in LNG. Nevertheless, certain assumptions can be made based on the physics of flow and behavior of particles in the regions where both gravity and Brownian motion have a significant contribution to the deposition rate. Apart from radon and radon progeny physical properties, it is important to possess information on LNG flow properties since the behavior of particles also depends on the fluid. It is assumed that LNG is a Newtonian fluid, i.e. the relation between the shear stress and the shear rate is linear, and viscosity (µ) is independent of strain rate. The same assumption was stated in several studies such as [78] – [79]. In this case, the equation for Reynolds number, which is the ratio of inertial force to viscous force, is:

$$Re = \frac{\rho V D}{\mu} \qquad (5)$$

Where:
ρ – fluid density (kg/m^3)
V – fluid velocity (m/s)
D – pipe diameter
µ - fluid viscosity (kg/(m.s))

With the help of Reynolds number value, it is possible to identify the type of flow, i.e. whether flow is laminar or turbulent. Laminar flow ($Re \leq \sim 2300$) is characterized by smooth movement of fluid layers whereas turbulent flow ($Re \geq \sim 3000$) is chaotic [80]. The types of flow are illustrated in Figure 9. Therefore, the type of flow will have a direct effect on the deposition rate of solid particles.

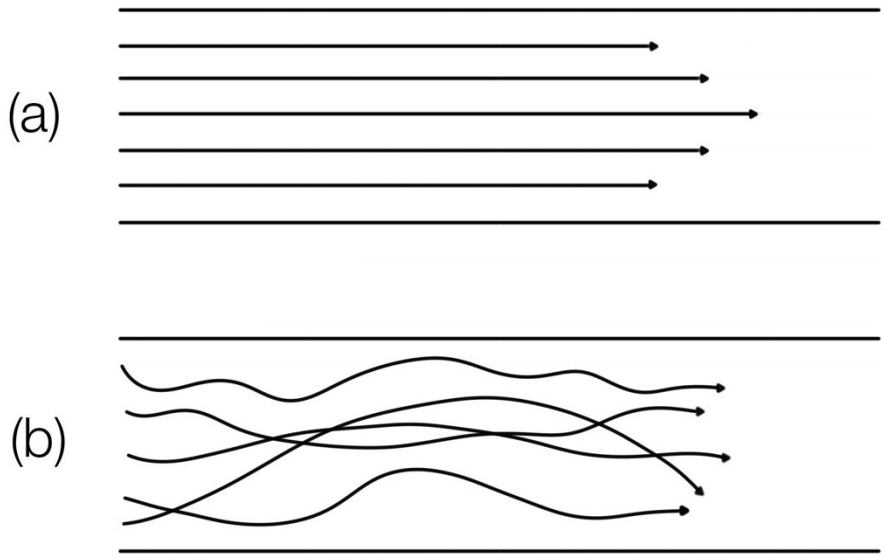


Figure 9 Two types of flow where a) is laminar flow and b) is turbulent flow [80].

As it can be seen from eq. (5), apart from physical properties of fluid such as density and viscosity, operation parameters such as a diameter of the pipe and fluid velocity also determine the type of flow. Therefore, the value of Reynolds number will differ depending on the operating conditions of different LNG installations. Nevertheless, according to several experimental data ([81] – [82]) Reynolds number of LNG flow is significantly high due to low viscosity of LNG, and value of up to 30,000,000 can be expected. Therefore, regardless of pipe diameter and fluid velocity, it can be assumed that in all LNG installations the type of LNG flow is turbulent flow. However, depending on the carried processes, for instance during LNG unloading [83] or LNG vaporization [84], a two-flow phase, i.e. gas and liquid, occurs. In this case, we are dealing with the complex system where it is even more difficult to predict the behavior of radon and radon progeny. For simplicity, it is assumed that in all LNG systems, there is a uniform fluid with the same properties. Due to the sophisticated system, it is challenging to predict how exactly radon and radon progeny will behave in the gaseous state of LNG and liquid state of LNG. However, as it has been mentioned above, LNG flow is turbulent flow, and the same is applicable to LNG in the gaseous state [85]. Therefore, it can be assumed that regardless of the state of LNG, radon and radon progeny will have almost similar deposition behavior. There is a note which is worth mentioning: in gaseous systems, for instance, in LNG regasification terminal, it is assumed that thermophoresis will facilitate the accumulation of radon and radon decay products on inner parts of equipment whereas in systems where LNG is mainly in the liquid state, thermophoresis will hinder the deposition process on the inner surface. The assumption will be discussed in detail in the following chapter.

2.4.2.1 LNG piping systems

Piping system is one of the systems where radon and radon decay products can be observed. As it has been mentioned, when it comes to submicron or nanoparticles, Brownian diffusion and thermophoresis are the main factors affecting the deposition

rate of the particles. Most of the studies are conducted on the thermophoresis phenomenon in hot gases in cold pipes, and there is a lack of knowledge on solid particle behavior in the cryogenic liquid. Nevertheless, some assumptions can be made based on the working principle of the thermophoresis and Brownian phenomenon and processes which are going on in LNG systems.

The main principle of the thermophoresis phenomenon is the following: the thermophoresis depends on the temperature gradient in the surrounding gas molecules. The gas molecules coming from the hot side have a greater velocity and energy compared to gas molecules coming from the cold side. Therefore, the greater gas molecules from the hot side will have greater momentum and push the particle in the opposite direction, i.e. to the direction of the cool side [86]. Such thermophoresis is called "positive". Nevertheless, there is also "negative thermophoresis" when solid particle moves from cold towards hot region. The mechanism behind negative thermophoresis is sophisticated and hasn't been studied in-depth yet. However, there is a study where radon daughters were collected via thermophoresis and electrostatic attraction in personal dosimetry in uranium mines [87]. In the study, positive thermophoresis is used in order to collect radon daughters. Therefore, it is assumed that radon and radon progeny tend to move from hot side to cold one. For radon and radon progeny in LNG flow the following scenario is assumed: even though LNG piping system is insulated, there is still heat leakage from ambient environment [88]. Therefore, the temperature closer to the inner walls of pipes is higher compared to the temperature further from the walls towards to middle of the pipe. The particle near the wall with higher kinetic energy will "kick" radon and radon decay products away from the walls of the pipeline. However, one should keep in mind that thermophoresis is especially applicable to the cases when the temperature gradient is high, i.e. when the temperature difference between the wall of equipment and fluid is large, like in furnace. For the pieces of equipment where LNG is in the liquid state, it is assumed that the temperature difference is not really high, and there is a possibility for radon and radon decay products to be attached to the inner walls of pipe due to the strong turbulent nature of LNG flow and Brownian motion. It is also assumed that thermophoresis plays much more important role in LNG systems where LNG is in gaseous state (NG) due to the fact that turbulence is lower compared to LNG, and a temperature difference between walls and flow of NG is higher. Therefore, in straight pipes with NG flow, it is assumed that there will be higher deposition of radon and radon progeny. However, the concentration in straight pipe should be less both in NG and LNG systems compared to other locations. It is assumed that there are certain locations along the piping system where radon and radon progeny can be observed with a greater probability. The locations with greater probability are elbows, valves, t-bends or t-joints, sudden expansion and sudden contraction regions. The main reason behind assumption is a change in numerous parameters such as pressure, velocity, vortices and other parameters in these locations, and aforementioned changes can facilitate the accumulation of solid particles.

The elbow of the pipe is considered as one of the locations where there is a probability of accumulation of radon and radon decay products. In the elbow, solid particles tend to separate from gas flow due to centrifugal forces. These particles tend to form a relatively dense phase structure in the elbow known as a rope. In the

location of rope, the concentration of solid particles is much higher compared to other areas of pipe cross-section [89]. The process is illustrated in Figure 10. The small-scaled model developed by [90] showed that the concentration of solid particles in the rope region is higher by a factor of around 10 compared to other sections of the pipe. According to [91], in the vertical-to-horizontal elbow, the velocity of solid particles at the bend exit equals to one third of the average gas velocity, and the solid particles accumulation was caused by frictional forces. For particles more than 20 μm in duct bends, accumulation is lower in horizontal-to-vertical downward orientation due to gravitational forces which leads to less solid particle deposition compared to horizontal-to-horizontal orientation [92]. Despite the fact that radon and radon progeny are submicron particles, due to high density, large particles can be affected by gravitational forces, so it can be applicable. The rope formed in horizontal-to-vertical elbow is due to the inertia phenomenon and secondary flow [93].

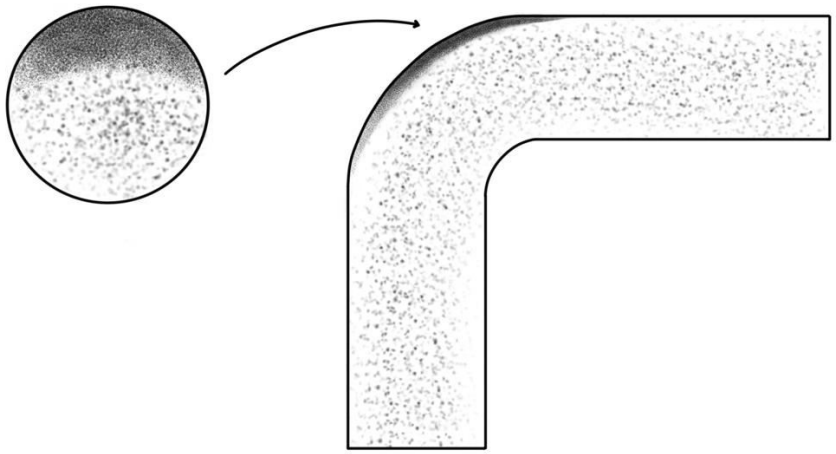


Figure 10 Rope formation and dispersion [89].

Nevertheless, aforementioned facts are applicable to the large particles due to inertia, and also to particles with high density. Therefore, it is applicable to radon and radon progeny particles with diameter exceeding 500 nm, or heavy particles. On the other hand, small particles with low inertia may follow the stream [94]. Therefore, for small nanoparticles where it can be assumed that density does not contribute to deposition rate and Brownian diffusion plays the major role, there can be different scenarios which should be examined.

The nanoparticle deposition in pipe or bend regions in turbulent flow has not been fully studied yet. There is a lack of experimental studies due to the fact that particle deposition behavior is complex and can be affected by numerous factors even in straight pipe, channels or ducts [95]. Nevertheless, according to the study [96], the minimum of deposition enhancement factor for nanoparticles occurs near the inside edge, and the maximum occurs near outside edge implying that outside edge will have more deposited radon and radon decay products. This actually corresponds to “rope” phenomenon mentioned above. In addition, in the case of higher Reynolds

number, the nanoparticles take a shorter time for deposition. Thus, it can be assumed that elbow is the region where relatively large and small particles as well as heavy ones tend to deposit.

Other areas in piping system where it is assumed that the accumulation of solid particles can be observed are locations around valves. In the study where computational fluid dynamics (CFD) modelling was used in order to investigate particle deposition in natural gas pipelines, correlation between ball valve opening and solid particles deposition ability was observed [97]. As the ball valve opening angle decreases, more energy loss or pressure drop can be observed due to increased vortices, which, in turn, leads to more solid particles deposition. Due to the pressure drop, the temperature of fluid decreases leading to retrograde condensation and formation of nuclei particles. Due to the turbulent flow, these particles are brought together and coagulation phenomenon occurs. Since radon and radon decay particles are already in solid state, condensation phenomenon is not applicable to radon and radon progeny in LNG. Nevertheless, it can be assumed that due to pressure drop and energy loss, as well as turbulent flow which makes radon decay products stick to the inner wall of the pipe as it has been mentioned before, downstream locations of valves are of particular interest.

Sudden expansion or contraction are also regions where accumulation of radon and radon decay products may occur. In the location of sudden expansion, recirculating flow occurs where submicron particles can be captured and deposited [98]. According to study [74], which has been already mentioned, radon and radon progeny are submicron particles. Therefore, it is assumed that both unattached and attached radon decay products can be captured by recirculating flow. The same approach is applied to sudden contraction with contraction angle more than 60° since at angles less than 60° recirculation zone does not occur [99]. Nevertheless, there are various parameters which affect deposition efficiency in sudden expansion and contraction areas such as Stoke number, contraction ratio, contraction angle and other associated parameters.
In general, for piping system, any region where LNG flow changes direction, including t-joints [100] or bifurcating tubes [101], is considered as region where there is a high probability of radon deposition on the inner part of walls. Apart from locations which have been noted, it is also assumed that radiation can be detected near pumps since, as it has been already noted in previous chapters, significant NORM contamination of inner parts of pumps was observed in certain gas processing units.

When it comes to LNG regasification terminal which is another type of land-based system, LNG vaporizer or ambient air vaporizers (AAV) is assumed to contain some amount of radon decay products. LNG flows into the AAV fin tube and is converted to natural gas, i.e. regasification process occurs. At the same time, the temperature of the fin tube of AAV and the ambient air is decreased, and this results in frost formation as the operation time lasts. Temperature and partial pressure of water vapor are the main factor causing the frost formation. Water vapor is either condensed first and then freezes into frost crystal, or water vapor is converted into frost crystal directly [102]. As LNG is converted to natural gas in AAV, the temperature rises from around 114 K to the temperature which is slightly lower than

ambient air temperature. In addition, it was also observed that the frost layer of AAV inlet is thicker compared to the outlet [103]. Based on the provided information, several assumptions can be made:

I. **Radon and radon decay products can be observed in frost formation.** Radon and radon decay products tend to attach to different substances including water vapor [104]. Therefore, it is assumed that radon and radon decay products can deposit on inner parts of AAV and constitute a part of frost formation. Nevertheless, it is assumed that there is a higher possibility to observe radon decay products on inner parts of AAV rather than radon due to the fact that the state of radon changes to gaseous one whereas radon daughters stay in solid state.

II. **Radon tends to concentrate in the inlet of AAV.** Due to the fact that temperature is rising from inlet to outlet of AAV, radon tends to vaporize together with LNG due to its boiling point which is around -62°C [65]. Therefore, it is assumed that radon tends to accumulate near inlet locations where it is still in solid state rather than in outlet.

Other pieces of equipment of LNG regasification installation such as BOG recondenser and compressors can also contain radon and radon progeny.

LNG storage tank is assumed to be another location of radon and radon decay products accumulation. It is assumed that in LNG storage tanks sedimentation or settling process takes place: in still environments, under gravity, heavy solid particles tend to deposit on the bottom of the tank. Nevertheless, due to the external heat leakage to the tank, there is convective circulation of fluid close to LNG walls [105]. It is assumed that this recirculating flow causes small and light radon and radon progeny particles to be stuck to the wall, and relatively larger particles are captured by the flow and remain in the layers near walls of LNG fluid. Apart from recirculation flow, rollover phenomenon which occurs in land-based LNG storage tanks may lift radon and radon decay particles which were deposited via gravitational force from the bottom due to the motion of fluid layers. Thus, it is assumed that there is always radon and radon decay products presenting in LNG. There is a record on NORM deposition in NGL storage tanks, where it is stated that NORM is deposited due to long hold up times [60]. The same logic can be applied to LNG storage tanks. Nevertheless, even though radon decay products concentration will increase, substantial amount of radon can be decreased due to the half-life of radon which is 3.8 days [10]. Apart from storage duration which affects radon concentration, transportation to desired site should also be taken into consideration. Currently, there are two LNG receiving terminals in Sweden: one in Nynäshamn, and one in Lysekil. LNG for both terminals comes mainly from Gasum LNG plant located in Risavika, Norway. Sweden also receives LNG from Gate LNG terminal located in Netherlands and Zeebrugge LNG terminal located in Belgium. LNG containers from Gate and Zeebrugge LNG terminals reach Gothenburg in five days. However, both Gate and Zeebrugge LNG terminals receive LNG from different sources such as Qatar, US or Russia; therefore, initial LNG is transported for more than five days before reaching Gothenburg [106] – [107]. It implies that radon concentration will actually decrease before reaching

Gothenburg. However, most likely, radon concentration in LNG which comes from plant located in Risavika will remain almost the same due to shorter transportation time. Therefore, different LNG terminals or different sites, like Gothenburg, to which LNG is transported, will have different radon and radon decay products concentrations based not only on natural gas source but also on LNG transportation period. Thus, NORM contamination of particular system is the sum of different factors and variables, and it is challenging to predict NORM contamination level in advance.

There is a very low probability of radon and radon progeny deposition in engine room. Due to the fact that alpha and beta particles cannot penetrate the equipment, it is assumed that there is a very low probability of detecting alpha and beta radiation in air in the engine room. Level of gamma radiation can be present since gamma radiation can penetrate. However, since many parts of LNG systems are made of nickel steel, and certain thickness of steel is able to block gamma radiation to some extent, it is assumed that gamma radiation poses little or almost no hazard. In addition, due to the fact that strong gamma emitters are short-lived radon daughters with half-life between 160 microseconds and 27 minutes, it is less likely that one will be able to detect their presence at site of LNG fuel systems where radon is unsupported. In addition, there are LNG spill detectors installed on ships which help to identify the escape of LNG out of equipment. In the case of escape, alarm turns on, and needed procedures in order to prevent the leakage are applied. Therefore, it is assumed that in normal operational conditions, i.e. in the absence of leakage or the presence of systems which are able to warn about undesirable leakage, there is no risk or very low probability of radon and radon decay product presence in the air of the engine room.

The region where alpha and beta particles can be detected outside of equipment is near flanged connections due to LNG leakage [108]. Therefore, flanged connections are regions with higher probability of alpha and beta radiation presence in air due to possible leakage. Nevertheless, ideally, there should be no leakage. Therefore, even though flanged connections and close regions are considered as regions where there is a higher probability to detect alpha and beta radiation compared to other regions, there is still a low overall probability of alpha and beta radiation presence.

Overall, LNG storage tank, namely both land-based tank and ship tank, is the piece of equipment where there is the highest probability to observe and detect the presence of radon progeny due to long hold up. Apart from LNG storage tank, on inner parts of piping system, especially outer edge of elbows, valves, t-joints, sudden contraction and expansion, and any region where direction of LNG flow changes, there is a probability of radon and radon decay products deposition. Pumps also should be checked for NORM deposition. In general, LNG ship tank and land-based tank should be especially investigated, and samples for investigation from inner surface of LNG tank preferably shall be taken. As it has been noted, the engine room is the location where the probability to detect radon and radon decay products is low, it is also assumed that there is no alpha and beta radiation due to inability to penetrate the pieces of equipment and due to the absence of leakage spots. There can be leakage spots in engine room but they should be detected, so, ideally, it is considered that there are no such spots from where radon and radon

progeny can escape. The inner parts of LNG fuel system shall be investigated to ensure safe maintenance procedure.

2.4.3 Maintenance procedure and radiation exposure

The maintenance should be carried twice within a five years period on LNG cargo containment systems. The typical lifespan of LNG cargo systems and LNG facilities is beyond 20 years [109]. It implies that workers might be subjected to radiation at least 8 times. There are also other possible ways of being exposed to radiation such as gamma radiation exposure from short-lived radon daughters, inhalation of radon and radon decay products from combustion of natural gas as well as from possible leaks [110]. In order to measure radiation damage to tissue, the concept called absorbed dose is used. The absorbed dose shows the intensity of the energy deposited in a matter per unit of mass, and the unit of absorbed dose is called gray (Gy). Nevertheless, each type of radiation has its own damage weight which is important to take into consideration when one attempts to evaluate the damage which has been done to tissue. Therefore, the concept of equivalent dose has been introduced which is the absorbed dose multiplied by a radiation weighting factor. The tissue affected by alpha particles receives more damage compared to the tissue from beta radiation. For instance, for gamma rays, X-rays and beta particles, the factor equals to 1 whereas for alpha particles the factor is 20. The unit both for equivalent dose is the sievert (Sv) [45]. The sievert is also the unit for effective dose which is, as defined by [111], *"summation of the tissue or organ equivalent doses, each multiplied by the appropriate tissue weighting factor"*. Therefore, the sum of the weighted equivalent doses is effective dose. The effective dose concept is widely used in radioprotection since it determines how detrimental and hazardous the exposure to radiation is. In addition, effective dose also shows how much dose was absorbed by all organs of the entire body, as well as the sensitiveness of each part of the body to radiation. Taking into consideration aforementioned facts, it is essential to evaluate the risk associated with the presence of radon decay products in LNG and LNG fuel systems.

International Atomic Energy Agency (IAEA) published International Basic Safety Standards for Protection Against Ionizing Radiation and for the Safety of Radiation Sources where recommendations on three types of situation according to International Commission on Radiological Protection (ICRP) can be found, namely planned exposure situations, existing exposure situations and emergency exposure situations. According to the document, for workers who are over the age of 18, the dose limits are [111]:

I. A dose of 20 mSv annually averaged over five consecutive years making 100 mSv in 5 years, or 50 mSv in a single year.
II. A dose to the lens of the eye of 20 mSv annually averaged over 5 consecutive years making 100 mSv in 5 years, and of 50 mSv in a single year.
III. A dose to the extremities (hands and feet) or to the skin of 500 mSv in a year.

Taking into consideration all aforementioned facts, especially regarding directives applied by Basic Safety Standards Directive to all European Union (EU) member states and due to Systematic Work Environment Management guidelines developed by the Swedish Work Environment Authority, the concentration of radon in LNG fuel systems should be measured. It can be done by conducting experiments and evaluating obtained data.

2.5 Radon and radon progeny measurement

Different ways of measuring radon and radon progeny concentration are proposed in this study. In the first place, a theoretical background will be provided. Due to the fact that radon and radon progeny will be measured not only in natural gas but also in engine room, techniques and comprehensive review of radon measurement both in air and natural gas are provided.

2.5.1 Introduction

Radon concentration measurement in air is much more common compared to the measurement of the radon progeny concentration. Nevertheless, in the case when a more accurate estimate of the radiation exposure is required, radon decay products are analysed even though the measurement procedure is more sophisticated and costly compared to the one of radon itself. Sometimes, both radon and radon progeny levels are measured, and one of the measurement ways is the use of the two-filter method. The two-filter detector operates in the following way [112]:

I. The air where both radon and radon daughters are present passes through the first filter where radon progeny is left on the filter whereas radon itself travels further.

II. After radon progeny is removed via the first filter, radon together with sample air enters a decay chamber, where the daughters produced are collected on a second filter.

III. The concentration of radon progeny and radon can be evaluated separately by different techniques such as alpha spectroscopy, scintillation counting, or a passive detector.

Apart from different measurement techniques that will be discussed in depth further, there are three different time regimes for sampling, namely instantaneous, or “grab” sampling; continuous, or “sniffing” sampling; and integrated sampling.
Instantaneous sampling has several advantages such as prompt delivery of results and the ability to easily carry equipment due to the fact that used equipment is usually portable. However, there are several disadvantages, and one of them is the need of result correction for the presence of Rn-220. The precision of grab sampling is relatively low unless counting times on the order of several hours are used. During the grab sampling, air is collected in the container which can be ranged between 5 and 15 litres, and carried to the laboratory for further investigation. In order to perform more accurate radon analysis, air with radon can be captured via activated charcoal trap submerged in liquid nitrogen. At this low temperature, radon remains on the trap. Afterward, the trap charcoal is warmed in the presence of helium, and the concentration of radon is analysed using a counting cell or ionization chamber [112].

In continuous sampling, first, the sample air passes through a filter so radon decay products can be removed. Afterward, the sample air goes through a sensitive volume where the analysis can be conducted. In the case if active system is used, the air is drawn through the sensitive volume with the help of a pump; however, in the case of passive system, the air diffuses into the volume. The latter process requires around 4 hours in order to reach equilibrium state. Compared to instantaneous sampling method, continuous sampling has relatively good precision. Some of continuous sampling devices can track hourly fluctuations of radon levels. The continuous tracking of radon levels can help to indicate poor ventilation and other vital parameters. The sampling period can range from 48 hours to months in the case if the filter is maintained or replaced on a regular basis. Continuous sampling usually requires trained staff: one of the reasons behind is due to the fact that devices should be calibrated annually, with semi-annual rechecks. In order to obtain a real picture of radon concentration, continuous monitoring should be conducted in the absence of dusty, smoky, and high-humidity environment. In the case if continuous monitoring is performed in an undesirable environment, it should be carefully maintained [112].

Integrated sampling gives a single average concentration over the measuring time. The length of the period varies starting from a few days to years. In order to perform integrated sampling, both active and passive devices are used; nevertheless, the use of passive devices is the most common [112]. There are numerous advantages of integrated sampling over other types of sampling methods. For instance, integrated sampling methods are less expensive compared to continuous sampling. In addition, integrated sampling devices can be used for radon levels that are too low for continuous and grab sampling methods. In addition, obtained samples can be analyzed later, at a more convenient time and place [113]. The most common type of integrated sampling device is integrating dosimeter. There are two types of testing periods: long-term or short-term [112].

I. **Short-term testing.** There is a wide range of short-term testing devices starting from the ones which should be deployed for 2 days to 3 months. As it has been mentioned above, for integrated sampling method, mostly passive devices are used among which the most common ones are alpha track detectors, charcoal canisters, charcoal liquid scintillation detectors, and electret ion chambers. Due to the fact that radon concentration depends on numerous factors such as season, ventilation, and lifestyle of residents, a short-term test will more likely demonstrate a peak or valley of radon levels which in reality does not represent a complete picture of radon levels throughout a year. Therefore, the majority of public health agencies opt for long-term test in order to understand whether some actions are required. Nevertheless, short-term test is preferred for initial sampling, and in the case if the radon levels exceed the acceptable standards, long-term test is used for more representative results.
II. **Long-term testing:** Long-term tests, unlike short-term tests, are utilized for more than 3 months. The most common devices which are used in order to perform long-term tests are alpha track and electret ion chamber detectors.

As it has been mentioned above, a long-term test is better for obtaining an annual mean value of radon concentrations.

There are two types of radon detectors: passive and active. Both types will be thoroughly reviewed in the following chapter.

2.5.2 Passive detectors for radon measurement in air

2.5.2.1 Activated Charcoal Detector (ACD)

Activated charcoal detector (ACD) is usually comprised of a cylindrical canister which contains activated charcoal. The charcoal is used in radon concentration measurement due to the fact that the charcoal has adsorptive properties, and radon can be captured on active sites of charcoal. Prior to measurement procedure, the canister should be prepared. When measurement procedure should take place, canister is unsealed. The exposure time usually varies between three and four days [112]. Nevertheless, it can be also between 2 and 7 days [114]. During measurement, radon enters via diffusion mechanism and is absorbed by charcoal. After the exposure period is over, the top of canister should be replaced, and canister itself should be sealed. The level of radon can be analyzed with the help of different devices. One of the most common devices is NaI(Tl) scintillation detector which helps to measure the intensity of the gamma radiation generated by radon daughter products, usually Pb-214 and Bi-214. Nevertheless, high purity intrinsic germanium (HPGe) semiconductor detectors can also be used. Another activated charcoal detection technique utilizes liquid scintillation counting method. One to two grams of charcoal for radon capturing is used, after which the charcoal is placed into an organic solvent in which radon is highly soluble. Afterward, aforementioned scintillation counting in order to detect alpha and beta radiation types is used [112].

The activated charcoal detector is considered as one of the most inexpensive devices for radon level measurement. In addition, there is no need for trained staff in order to carry the measurement; therefore, the ACDs are very popular for screening of private dwellings. Nevertheless, ACD has several disadvantages: the radon adsorption is temperature and humidity dependent process. Therefore, ACD should be used only in indoor environments where there are slight fluctuations of temperature and humidity [112]. Though there is a correction for humidity, there is no any correction factor for temperature. In addition, ACD has one more drawback: in the case if the concentration of radon experiences significant fluctuations, the obtained results may not depict the real radon levels. The reason behind is the equilibrium phenomenon, namely the shift between radon adsorption and desorption processes in the case of radon concentration change. This is particularly essential when there is a decrease in radon concentration closer to the end of the exposure since the amount of absorbed radon may decrease. This, in turn, leads to results which do not depict real radon concentration levels [115].

2.5.2.2 Electret Ionization Chambers

The electret ionization chamber is comprised of a conducting plastic chamber containing the electret [112]. The electret is the type of dielectric material containing quasi-permanent electric charge or molecular dipoles which are able to

create both internal and external electric fields [116]. The simplified cross-section of chamber is shown in Figure 11.

Radon enters to the chamber, more precisely through filter to the chamber, in the same way as it enters to ACD – via diffusion phenomenon. There, radon undergoes decay process, and at the same time ionize the air. The electrons generated during ionization process are drawn to the electret with further decrease of the potential difference, or voltage, on the surface of the electret. The voltage change is measured by a special calibrated voltmeter. The electret ionization chamber is used until the electret surface potential is depleted. Afterward, the electret should be replaced [112].

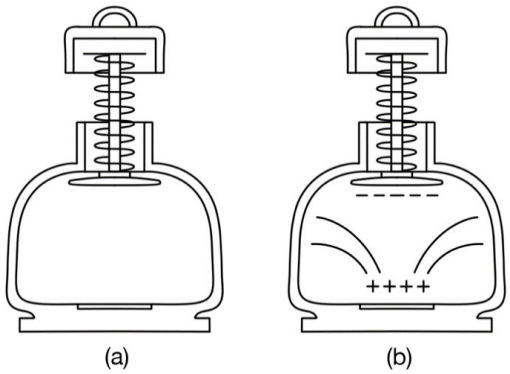


Figure 11 Simplified cross-section scheme of open S-type electret ion chamber. In (a) case the cross-section itself is shown while in the case (b) the principle of electret ion chamber is displayed. Electrostatic field attracts free ions, and positively charged electret is further analyzed for radon levels [132].

The electret ion chambers may be utilized both for indoor and outdoor environments due to the fact that they are less sensitive to the changes in temperature and humidity compared to other types of detectors. Electric ion chambers are suitable both for short-term and long-term measurements: in the case of short-term measurement, the thinner electret is used whereas for long-term measurement a thick electret is required. Nevertheless, though this type of detector is not sensitive to changes of certain environmental factors, electret ionization chamber is sensitive to all types of natural background radiation. Therefore, calibration is required [112].

2.5.2.3 Alpha Track Etch Detectors

The alpha track detectors are based on the following mechanism: alpha particles which were generated during the decay process of radon and radon progeny leave a so-called track of broken chemical bonds as they passing through particular solid materials. In order to make created tracks visible, some of these solid materials can be subjected to chemical etching. Cellulose nitrate (LR-115) and allyl diglycol carbonate (CR-39) films are two of the most commonly used materials [112]. For instance, CR-39, to be more precise its polymer, i.e. polyallyl diglycol carbonate (PADC), is consisted of carbon, hydrogen and oxygen atoms. When emitted alpha particles passes through the film, they collide with the electrons of oxygen atoms which leads to loss of the energy resulting in the positive ionization of oxygen

atoms in detector film. By applying strong acidic or basic solutions, track can become visible [117].

Despite the fact that the etching procedure can be performed both in acidic and basic solutions, mostly NaOH, i.e. basic solution, is used. Afterward, the visible track density is either analyzed by manual scanning with the help of a microscope or via a computer-based optical scanning program. One of the advantages of using alpha track etch detectors is that there is no need for ambient background correction. The reason behind is that the tracks which are left by beta and gamma radiation are not imaged by the etching, and only tracks left by alpha particles are visible after etching. Nevertheless, not all alpha particles are capable of creating the track: only the portion of the alphas with sufficient kinetic energy creates a track which is large enough to be imaged after etching. Other advantages are the cost and simplicity of this type of detector. The alpha track etch detectors are inexpensive and easy to deploy. Consequently, they gained the widespread popularity and are used for long-term measurements of radon levels. Compared to other detectors, alpha track etch is less sensitive: therefore, for more precise and accurate measurements, the large period, namely the period of 3 months to 1 year, is required [112].

2.5.2.4 Thermoluminescent Dosimeters

Thermoluminescence (TL), also known as thermally stimulated luminescence, is a process in which a solid, usually minerals in crystalline form, emits light by being subjected to a heat. The light is emitted due to the fact that the energy which was absorbed from radiation is re-emitted. Despite the fact that re-emitted light is usually a reason of previously absorbed energy from radiation, the excitation can also occur via pressure, friction, or light [118]. The mechanism of TL is the following: after material is being subjected to the radiation, both types of charge carriers, i.e. electrons and holes, are trapped in the forbidden gap. By applying a heat to the material to a few hundred degrees centigrade, either electrons or holes recombine with charges of the opposite sign, and recombination results in a visible light. Released light is nothing but previously absorbed energy, i.e. the radiation dose which material received. Due to the fact that thermoluminescent dosimeters (TLDs) are capable of recording all radiation doses, TLDs should be accordingly adjusted. The working principle of TLDs is the following: first, radon decay products are filtered from the ambient air. Afterward, the daughters from the radon decay tend to attach to the surface of the TLD. In order to identify radon concentration, the TLD should be removed from the detector assembly and placed in a reader. Here, in a reader, TLD is subjected to heat and the released light is measured. Several configurations of TLD exist. For instance, in one type of the detectors, the moisture first is removed with the help of silica gel. Afterward, the air passes through a filter that removes radon progeny, and then into a funnel-shaped chamber. In the chamber, with the help of electrical field generated by battery, the daughters tend to attach to the surface of the TLD [112].

2.5.3 Active detectors for radon measurement in air

2.5.3.1 Scintillation Detectors

Scintillation cells, or scintillation detectors, are widely used and established technology for radon detection. When radon undergoes decay process, alpha particle

is generated. The alpha particle hits the scintillation substance, and a portion of the kinetic energy is converted to a photon that can be detected by a photosensitive device such as a photomultiplier tube or a photodiode. The electrical pulse is generated, amplified, and can be analyzed in order to determine the radon concentration.

Different types of scintillation materials exist. The most common one as well as one of the earliest known material is silver activated zinc sulfide [ZnS(Ag)]. First, the air together with radon is drawn into the cell. There can be filter in order to eliminate radon progeny prior to interaction with the cell. The counting procedure should be done in the dark environment: alpha particles strike ZnS resulting in light which can be observed by photosensitive monitor. Apart from alpha particles, beta particles can also be identified; however, ZnS is not sensitive to gamma radiation. Silver activated zinc sulfide is also referred to as radon cell, or as Lucas cells, after the inventor of the original cell [112]. The Lucas cell (LC) detectors, which are among the most sensitive devices, are capable of detecting low radon levels [119].

Another type of scintillation substance which is also widely spread is sodium iodide (NaI) doped with an impurity (Tl). Unlike ZnS, this type of detector is sensitive to gamma radiation and insensitive to alpha and beta particles. The vessel which is used together with NaI should be made from the material which does not block gamma radiation. Therefore, such materials as aluminum or polyethylene which allow gamma rays to pass through are used. Nevertheless, NaI detector is also sensitive to surrounding gamma radiation. Therefore, additional shield made from lead or concrete, materials which do not pass gamma radiation, is required. The working principle of NaI scintillation device type is following: first, air containing radon is drawn into the vessel, and the vessel is put in the position with respect to the photomultiplier tube. Gamma radiation leads to light flash generation that can be further analyzed [112].

Cesium Iodide (CsI) crystal scintillator detectors can also be used. Another scintillator device, liquid scintillator, is also used for radon concentration measurement. The ability of radon to dissolve at low temperatures is actively used in liquid scintillator. Air containing radon passes through an organic solvent where radon is highly soluble. The remaining steps are almost the same as in all other types of scintillator: while radon is decaying giving away alpha and beta particles, generated light flashes are detected by a photosensitive detector and counted. Due to the fact that radon and radon progeny are closely mixed together with the scintillator substance, the counting efficiency of liquid scintillator is higher compared to other types of scintillator devices. However, liquid scintillator requires the use of wet chemistry which makes the procedure more sophisticated [112].

2.5.3.2 Internal Gas Ionization Chamber Counters

Ionization chamber is comprised of a chamber with two electrodes: anode and cathode. A voltage potential between the electrodes creates an electric field within the chamber. The scheme is shown in Figure 12. The radiation emitted during radon and radon progeny decay process ionizes the air molecules leading to generation of electrons and positively charged ions. Both electrons and positively charged ions move to the electrodes of the opposite polarity under the influence of the Coulomb

forces [112]. Different modes of operation for ionization chamber exist. One of the modes is a "current" mode: the electric field is strong enough for continuous process, where recombination of ion pairs can be avoided. Ionization chamber counter requires complicated measurement approaches due to the fact that the currents are small [120].

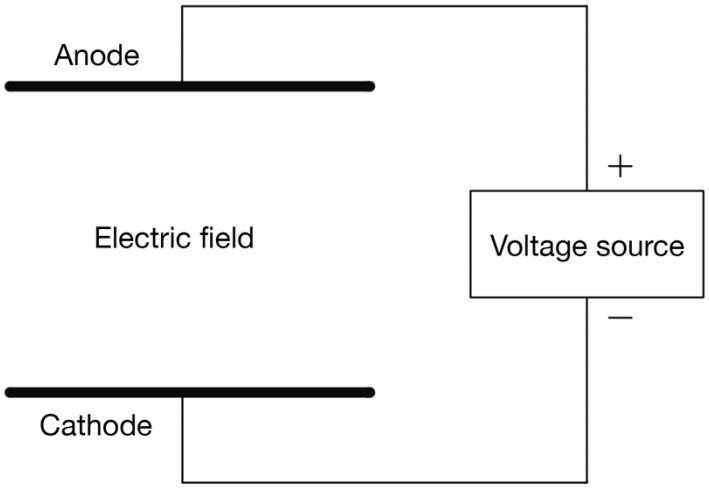


Figure 12 Simplified scheme of ion chamber principle [112].

Due to the aforementioned complexity, this type of detector is also among the most expensive ones. Nevertheless, ionization chambers are also very accurate and are considered as one of the most sensitive techniques for radon levels measurements [112].

2.5.3.3 Semiconductor Detectors

Semiconductor detectors, compared to all other types of detectors, are able to measure radon concentrations with the highest accuracy. Semiconductor detectors possess certain advantageous features such as *"high readout speed, the direct availability of signals in electronic form, the simultaneous precise measurement of energy and position and the possibility of integrating detector and readout electronics on a common substrate"* [120]. Most of devices are capable of distinguishing between the alpha radiation from radon and alpha radiation from radon progeny. In addition, these devices are also able to distinguish between radon and thoron, which is radon isotope [112].

The principle of working is similar to the one in ionization chamber counter. Semiconductor materials react to ionizing radiation: the particle which passes through the detector leads to generation of the electron-hole pairs. The number of the pairs corresponds to the energy which semiconductor received being exposed to radiation. Electrons and holes are received by electrodes in the presence of an electric field, and it results in an electric pulse which can be further analyzed. As it has been mentioned, the number of the pairs corresponds to the radiation energy; therefore, received pulse provides data on the radiation energy [112].

2.5.4 Measurement of radon in LNG

To our knowledge, there is no established method of radon level measurement in LNG. Nevertheless, there is a research conducted on measurement of LNG trace constituents including radon [11], where LNG was collected using cryogenic system. Later radon was collected using carbon cartridge based on EPA 520/5-87-005 [114] standard followed by radon concentration measurement via gamma ray spectroscopy. However, aforementioned techniques are not considered in this paper due to the following reason: in research [121], numerous flaws of EPA 520/5-87-005 standard were demonstrated. Therefore, obtained radon concentrations may not reflect the real radon concentration due to inaccurate algorithms. In addition, as it has been mentioned above, ACDs have numerous disadvantages which can have affect the results.

Therefore, in this study, the following technique is proposed: LNG should be converted to gaseous state following a certain standard such as ISO 8943:2007 [122], and the radon level in natural gas can be further measured with the help of different methods such as alpha scintillation cell and electret ion chamber. In order to identify the level of radioactivity of LNG equipment, the sample of material which was exposed to radiation, such as the sample from inner surface of LNG tank or any other type of equipment where radon concertation is assumed to be relatively high, should be taken in accordance with a certain standard such as ISO 7503, and delivered to the lab for further investigation. The wipe sampling can also be done according to the rules of certain organizations such as [123]. Apart from aforementioned techniques, for preliminary NORM identification, special and sophisticated though expensive equipment which is capable of identifying radioactive element at once, i.e. just by making device closer to the source of NORM, can be used. The principle of this device was demonstrated during the visit to the Swedish Radiation Safety Authority. The test was performed on Cs-137, where after closer approach, device identified the element as well as showed the concentration on the display. This device can be used during real measurement of radon and radon progeny on the areas of LNG installations. In addition, scintillation detectors such as NaI can also be used for initial detection, and then other techniques can be deployed for obtaining more precise data.
To obtain representative sample of LNG, the sample should be converted from liquid to gaseous state without partial vaporization since partial vaporization leads to the loss of certain LNG components. In the case of prior vaporization, for instance in the sampling line, such components as nitrogen and methane will vaporize first resulting in so-called pockets of gas in the liquid stream resulting in the non-representative sample rich in heavy components [124]. To avoid aforementioned problems, one should follow strict instructions and standards in order to obtain representative analysis. ISO 8943:2007 provides guideline on how to carry sampling of LNG. Two sampling methods, namely continuous and intermittent, are described in ISO 8943:2007. The selection of the method depends on certain preferences; however, less equipment with moveable elements is required for continuous LNG sampling system compared to intermittent sampling one. In addition, during continuous sampling method, the process of cleaning sample containers can be less complicated [125].

As it has been mentioned above, there are numerous procedures which should be strictly adhered to. For instance, during the sampling, the flow rate should be stable, i.e. the sampling procedure cannot be carried during initial start-up or before stopping due to disturbed flow rate. Sample probe itself should be placed in areas in the pipeline where the LNG is in a sub-cooled condition. The procedure of sub-cooled condition calculation can be found in ISO 8943:2007.

2.5.4.1 Radon level measurement via alpha scintillation cell

There are numerous researches conducted on radon level measurement in air via alpha scintillation cell. It is widely used and well-established method for radon concentration measurement [126]. However, alpha scintillation cells can also be used for measurement of radon concentration in natural gas. Moreover, due to the fact that natural gas is less dense and lighter compared to air, the efficiency of alpha scintillation cells can be increased up to 6-7%. The reason behind efficiency enhancement is increased path-length of radon alpha particles implying more frequent interaction with ZnS [127]. In research [127] on radon level measurement in natural gas following the steps below:

1. Certain volume of gas is supplied to cells and, after certain period of time, cells are sealed.
2. After equilibrium between radon decay products is reached, each cell was measured via alpha-scintillation detector.
3. The formula to calculate radon concentration (Bq/m^3) is:

$$C_{Rn} = \frac{C_s - R_b}{60 \times V \times E \times DK} \quad (6)$$

Where:
C_{Rn} – radon concentration
C_s – gross count rates (counts/min, cpm)
R_b – background count rates (cpm)
60 – conversion factor from Bq to disintegrations per minute (dpm)
V – cell volume
E – counting efficiency (cpm/dpm)
DK – decay factor from the time of collection to the midpoint of counting

4. In order to check minimum detectable activity, the following formula is used:

$$MDA = \frac{2.71 + 3.29\sqrt{R_b \times t_s[1 + \frac{t_s}{t_b}]}}{t_s \times E} \quad (7)$$

Where:
MDA – minimum detectable activity
R_b – background count rates (cpm)
t_s – sample counting times in minute

t_b – background counting times in minute
E – counting efficiency (cpm/dpm)

2.5.4.2 Radon level measurement via electret ion chamber

Electret ion chambers (EIC) are widely used for radon level measurement in air, both indoor and outdoor. Nevertheless, natural gas differs from air, and different approach is required. In research [128], electret ion chamber was used in order to measure radon concentration. In order to establish the measurement system of radon in natural gas, the following procedures were applied:

1. Air-tight system was utilized and modified: two valves from which natural gas can go inside and outside were added to the system.
2. Two EICs which are in the "on" position were added to the system together with the National Institute of Standards and Technology (NIST) radon emanation standard.
3. Radon was supplied when required.

In order to measure radon itself, the following procedures were applied:

1. Initial voltage of electrets of both EICs were recorded.
2. The flow rate of natural gas was around 20 litres per minute (LPM). This amount of gas is sufficient in order to displace the air in the system.
3. After certain manipulations, the flow of gas was supplied for around 2 minutes. Afterward, inlet valve was closed whereas outlet valve was kept open for 15 seconds.

The sampling was finished, and natural gas was trapped inside the jar at atmospheric pressure. After one to three days, EICs can be taken for radon concentration measurement.

In order to calculate the initial radon concentration (IRC) based on the information on the average radon concentration (ARC), the following equations were used:

$$ARC = \frac{TIC}{D} = IRC \int_0^D exp(-\lambda t)\, dt \qquad (8)$$

Where:
TIC – time integrated concentration (pCi-days/liter)
D – days
λ – decay constant of radon in one day (0.1814/day)

After certain manipulations, the initial radon concentration equals to:

$$IRC = \frac{(ARC \times \lambda D)}{1 - exp(-\lambda D)} \qquad (9)$$

2.5.4.3 Uncertainties of LNG sample and measurement systems

Two parts of LNG sampling system contribute to overall system uncertainty: the sampling procedures, i.e. the sampling and vaporization steps, and, for this paper, uncertainty of radon concentration measurement. For sampling system, generally accepted uncertainty value is ±0.30 %. For sampling and vaporization systems where no partial vaporization occurs, the uncertainty can be lowered to ±0.2 % [125].

In order to evaluate the uncertainty of the whole process including sampling and radon concentration measurement, all uncertainties should be calculated based on error propagation equation:

$$\sigma_o = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \ldots \sigma_n^2} \quad (10)$$

Where:
σ_o – overall uncertainty

2.5.4.4 Uncertainties associated with alpha scintillation cell method

The uncertainties associated with alpha scintillation cell methods are the following: the one is due to the difference in pressure between sampling location and laboratory location, and another one is due to the different fillings, i.e. air and natural gas.

Due to the fact that cell can be filled at the location with one elevation, or one pressure, and the same cell can be refilled at different location with different pressure value, the error might occur due to the difference in air density. There is inverse relationship between path-length of radon alpha particles which has been mentioned before and pressure, or density, which can be demonstrated by the following equation:

$$R_l \rho_l = R_h \rho_h \quad (11)$$

Where "l" denotes low altitude and "h" high attitude. According to eq. (11), values for path-length of radon alpha particles at low elevation locations with higher air density will be smaller, and vice versa. In research [129], the difference in path-length of alpha particles at different elevations was demonstrated. Colorado Springs is located on higher elevations compared to Dayton, Ohio. The data can be observed in Table 4.

Table 4 Different values of radon alpha particles path-length based on location [129].

Radionuclides	**Dayton, Ohio**	**Colorado Springs**
Radon-222	4.12 cm	4.99 cm
Polonium-218	4.70 cm	5.70 cm
Polonium-214	6.70 cm	8.13 cm

In order to compensate the uncertainty caused by difference in altitude, the eq. (12) should be used [130]:

$$\%Error = \frac{(N_1 - N_2)}{N_2} \times 100 \quad (12)$$

Where:
N_1 – net counts at the pressure of location where cell is calibrated (reference pressure)
N_2 – net counts at the pressure of second location

Second error, as it has been already mentioned, is due to natural gas filling instead of air. The uncertainty in this case can be calculated in the following way:

$$\%Error = \frac{(N_{air} - N_{methane})}{N_{air}} \times 100 \quad (13)$$

Where:
N_{air} – net counts in air at a certain pressure
$N_{methane}$ – net counts in methane at the same pressure

The correction factor for scintillation cells which is equal to 1.07 was found experimentally [128]. The calibration factors for air were used for the measurement of radon in natural gas. These results should be divided by a correction factor *f* in order to obtain appropriate results. The following procedures were applied to determine a correction factor:

1. The air containing no radon is bubbled through a NIST radium standard solution. The sample was placed to scintillation detector and inspected further for radon levels.
2. The same procedure as in first stage was applied; however, instead of air, the natural gas containing no radon was bubbled, and radon concentrations via scintillation cell was analyzed.
3. The *f* factor was calculated by dividing the results obtained in the second step to the first.

The result obtained for *f* factor corresponds to the ratio of air density to the density of natural gas at atmospheric pressures and room temperatures.

2.5.4.5 Uncertainties associated with electret ion chamber method

EICs, like scintillation cells, are also dependent on the density of the matter. Nevertheless, unlike scintillation cells, higher densities increase the response of radon whereas lower densities decrease the response [128].

EICs which are utilized for x- or gamma-radiation measurements require correction in the case if EICs are used at atmospheric pressures different from the atmospheric pressure at which EICs are calibrated. At lower pressures, or in other words, at

higher altitudes, due to the presence of less number of air molecules, the less number of ion pairs in a given path length are generated, whereas at higher pressures, i.e. lower altitudes, more ion pairs are produced. Therefore, the response of EICs is dependent on the density of air: EICs will give lower response at higher elevations. The correction factors and pressure ratios are in direct relation [131]. However, proportionate pressure ratios are not applicable to alpha radiation, and in the case of radon and radon progeny decay, beta and gamma radiation ionization is lower than 1%, i.e. the ionization is caused primarily by alpha radiation [132]. In research [132], three different volume types of E-PERM chambers, namely L, S, and H chambers, and two sensitivities, namely short-term, or high-sensitivity, and long-term, or low-sensitivity, were tested on the effect of elevation. Elevation and related correction factors are demonstrated in Table 5.

Table 5 Correction factors for different types of E-PERM chambers at different elevations [132].

Elevation (m)	L-type chamber	S-type chamber	H-type chamber
0	1.00	1.04	1.03
305	1.04	1.02	1.04
610	1.10	0.97	1.00
915	1.14	1.01	0.98
1220	1.19	1.03	1.01
1525	1.23	1.07	1.05
1830	1.28	1.17	1.03
2134	1.32	1.22	1.05
2440	1.37	1.27	1.04

For L type of chamber, correction factors are substantial at an elevation starting from 300 meters. In the case of S chambers, correction factors are substantial at elevations above 1200 meters. For H chambers, it was found that there is no need for correction factors up to 2500 meters [132]. The authors of [132] research drew the conclusion that the correction factors are substantial in the case if *"the dimension of the chamber is smaller than the range of alpha radiation."*

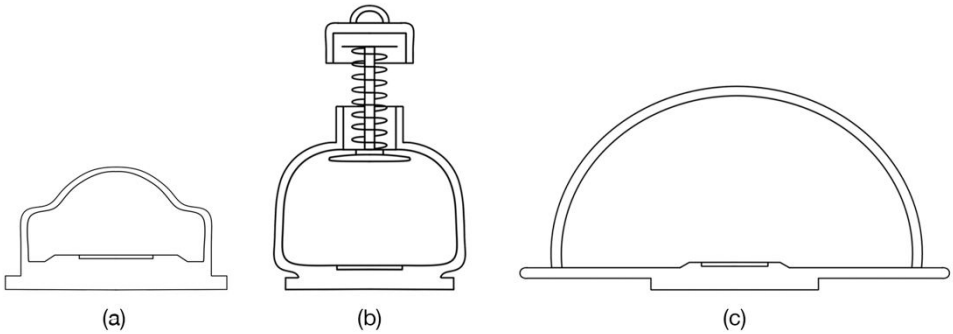


Figure 13 Simplified cross-section scheme of (a) L chamber, (b) S chamber, (c) H chamber [132].

The ionization energy, or potential, differs for air and natural gas. The ionization energy is the energy which is required in order to remove an electron from the atom.

In the case if ionization energy of a matter is lower than that of air, it results in generation of more ions by receiving the same alpha energy. Therefore, since natural gas has lower ionization potential compared to air, more ions will be generated leading to an over-response of EIC. The over-response of EICs was found to be 1.15. The correction factor *f* for EICs is almost the same as for scintillation cells – 1.10 [128].

2.5.4.6 Metal surface wipe sampling of LNG equipment

Wipe sampling is a method which helps to identify and estimate the possible contamination of the examined area [133]. In order to obtain representative results, it is of high importance to adhere all necessary procedures in right manner. For representative results, representative sample should be taken. In this study, it is suggested to follow the steps described in [123], which are, in turn, based on National Institute for Occupational Safety and Health (NIOSH) 9100 "Lead in Surface Wipe Samples" methodology. Certain assumptions which were developed based on the advice provided in [123] are also provided in this study.

Prior to starting the wipe sampling itself, it is highly recommended by [123] to follow some precautions such as assessing the area and location of testing on possible hazards as well as wearing personal protective equipment (PPE). Following items are included in PPE, namely: gloves, protective clothing, shoe covering, respiratory, and safety glasses. Nevertheless, in the case of wipe sampling of LNG equipment, not all PPE items may be required during wipe sampling. For instance, protective clothing should be used in the case if there is a certain contact with contaminated area.

It is assumed that a performed test, for instance, on the inner surface of LNG tank or spare equipment such as pump, may have a hazard related to the contact of body with contaminated equipment. It is assumed that there is no direct contact between contaminated equipment and body but rather a hazard related to beta radiation since contaminated surface can be a source of betta radiation due to the presence of long-lived daughters such as Pb-210 and Bi-210. Beta particles have the ability to travel a few meters in the air [134], and, as it has been mentioned before, beta radiation can penetrate a skin and be a cause of skin irritation. Therefore, in order to prevent any adverse effect from beta radiation, it is suggested to use protective clothing since beta particles can be stopped by a piece of clothes. Apart from protective clothing, during test procedure on any piece of NORM contaminated equipment, it is highly recommended to use gloves. The reason behind the recommendation of glove usage is the same as the reason behind the suggestion to use protective clothing – beta radiation. In [123], different types of glove materials are suggested such as nitrile, polyvinyl chloride (PVC), and natural rubber. In the case if sample is taken from the surface of LNG tank, which implies that a person is supposed to enter the tank, it is also suggested to have a shoe covering. According to [123], shoe covering is used in the case if the contact between shoes and contaminated cover is expected. Due to the fact that there is a probability of NORM contamination of the bottom of LNG tank, shoe covering is a PPE item which should be taken into account prior conducting a wipe sampling. Respiratory protection is also highly recommended due to the fact that inhaled alpha particles which can be emitted from the inner

surface of contaminated equipment can pose a hazard. Safety glasses should also be used since beta radiation causes the damage not only to the skin but also to the eyes.

After all precautions are taken, the wipe sampling can be conducted. Different types of equipment can be used for wipe sampling. For instance, sample can be placed in vial made either from plastic or glass, or it can be placed in the plastic bag with "zip" type seal. Sample media can also differ: gauze, paper, or pre-moistened wipe can be utilized during the wipe sampling process. Different metals require different types of solvents. According to [135], for certain type of radioactive element measurement method such as liquid scintillation counting, it is important to keep radioactive elements in solution. The contaminants are commonly divided between the ones which are soluble in water, and the ones which are soluble in organic solvents. For lead, both types of solvent, i.e. water and organic solvents, are applicable. In this case, it is suggested by [123] to focus rather on the surface material from which sample is supposed to be taken. Since LNG equipment is commonly made from steel or nickel steel, it is suggested to use solvents which will not damage aforementioned materials. However, it is also essential to take into consideration the method with which further sample will be analyzed. Therefore, in the process of selecting the solvent, both factors should be taken into account. The template can be either a plastic sheet or a cardboard, and suggested size by [123] is one square foot or around 929 cm^2.

The sample can be collected in different ways; nevertheless, in this study, NIOSH method is suggested and described. In order to collect the sample, following procedures should be adhered [123]:

1. In the first place, the wipe should be moistened in appropriate way since the wipe which contains too much solvent may cause the dilution of the sample. Therefore, the wipe with around 80% moistened area will be adequate and sufficient.
2. The application of wipe on the surface is shown in Figure 14. In (a), the first wipe, the direction of wiping, which is smooth up to down and the same smooth down to up, is illustrated above, whereas the circle illustrated below is the pad. In the second wipe, (b), the direction is from left to right and right to left, and the half of the pad illustrated below should be folded. In (c), the direction as in stage (a) is repeated, and quarter of the pad should be folded. In last stage, (d), pad should be folded again and put in the bag, which is illustrated, or vial, and labelled. In all stages, exposed side of the wipe should be folded in inward direction.
3. After the procedure is done, the wipe with contaminants on it should be placed either in a plastic bag or vial. Both plastic bag and vial should be sealed. The sample should be recorded, and this record identification should be noted on the bag or vial.
4. Afterward, PPE should be taken off in appropriate way, and disposed or cleaned accordingly. Other used materials should be treated in appropriate way as well.

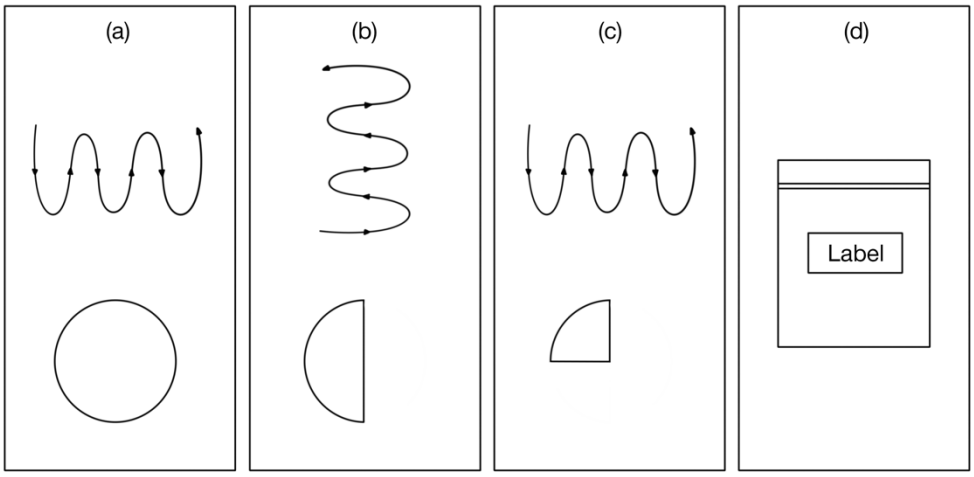


Figure 14 Representation of NIOSH surface wipe method [123].

For more representative results, it is suggested to take several samples [123]. For instance, in the case if samples are taken from the inner surface of LNG tank, it is suggested to take three samples from each surface, i.e. three from floor and three from wall. In addition, in the case if samples are taken from the different surfaces of LNG tank, it will be able to identify where radon and radon progeny tend to deposit more, and this will help to understand better the nature of radon and radon progeny behaviour in LNG. For other types of equipment, for example pump, due to the fact that surface area is not large, one sample will be sufficient.

2.5.4.7 Sampling of scale deposits in pipeline and NORM measurement

The information provided in this chapter was obtained during the visit to the Swedish Radiation Safety Authority.

First and foremost, the taken scale sample should be homogenized. It can be done with the help of manual gridding; however, the use of ball milling machine is more common technique to homogenize the sample. In the case if the taken sample is sludge, then it should be dried, and afterward, the same procedures as for scale sample should be followed. After sample is homogenized and put to the container, gamma spectroscopy can be carried. Liquid nitrogen is used in order to cool the crystal inside the device, for instance germanium crystal. Gamma spectroscopy is a technique which is more preferred compared to the use of alpha detector. The reason behind is complicated process of sample preparation, as well as elevated cost of the method due to the labour cost since the usage of radiochemistry.

3 Radon level measurement

3.1 Rapidos

Rapidos is an accurate short-term passive radon detector which can measure radon concentrations as low as 50 Bq/m^3 up to 150,000 Bq/m^3 over a week. The accuracy of Rapidos is 10% at 50 kBqh/m^3. The measurement is secure, and Rapidos detector is accredited according to ISO 17025 standard and approved all over Europe [136].

Rapidos is constituted of anti-static plastic material with a film inside. The alpha track technique is used for radon level measurement, and radon enters to Rapidos via diffusion. The detector material is poly allyl diglycol carbonate or CR39/PADC [137]. The working principle of alpha track etch detector was described in previous chapters. For precise measurement, Rapidos should be put indoor at a distance of at least 25 cm from floor, walls and ceiling, and 1.5 m away from air supply, external door and windows.

3.2 RadonEye

RadonEye is a smart radon detector with a high sensitivity (0.5cpm/pCi/l). RadonEye is the detector with high accuracy of <10% at 10pCi/l which can provide reliable data in a short period of time, usually less than one hour. It can be connected to the phone via Bluetooth where radon fluctuations can be easily observed. RadonEye employs pulsed ion chamber [138]. The ionization chamber detector is a type of alpha detector which is able to measure radon levels. The working principle of ionization chamber is following: alpha particles ionize different types of molecules which are present in air. These molecules alike alpha particles can travel longer distances with the help of air flow. These molecules help to identify radon level through the electrostatic detection of individual pulses from radon disintegration at a low level [139]. Measuring the average of radon level during the day does not depict the real picture since radon level varies depending on the time of day. In the study which was carried in numerous dwellings in order to check radon level during night and day, it is stated that the behavior of the inhabitants affects the radon and radon daughters' concentration during day. When inhabitants are sleeping or outside, windows are closed; therefore, ventilation is very weak which leads to higher radon concentration compared to day time since radon does not leave the house. Nevertheless, during day time, when inhabitants are home and open windows, there is good ventilation resulting in lower concentration. In the case when inhabitants are not home, the same logic as during night time is applied: windows are closed, so radon concentration increases [140]. Therefore, it is important to obtain information on a real-time radon concentration. RadonEye provides a real-time measurement, and in the case if radon level exceeds 148 Bq/m^3 (4pCi/l), alarm will sound [138]. However, due to the fact that engine room is always ventilated, it is assumed that radon level is almost the same during the day. It is also assumed that possible radon and radon daughters' presence in engine room is the highest while LNG fuel systems are deployed, and the lowest when LNG systems are off.

3.3 Radon level measurement process

Radon level measurement was carried with the help of aforementioned devices, namely Rapidos and RadonEye. The main objective of radon level measurement is comparison between above-mentioned pieces of equipment. The data on radon level was gathered and sent to the Swedish Radiation Safety Authority for further investigation, and with the help of this experiment, both devices can be thoroughly compared which will be extremely useful for real radon measurement in engine room and LNG facility areas.

There are six units of RadonEye and four units of Rapidos which can be seen in Figure 15. Each unit has its own reference number.



Figure 15 Six units of RadonEye, and four units of packaged Rapidos.

Before starting the measurement, it is of high importance to check whether there is an access to electricity since RadonEye requires the access to constant electricity supply. Besides, the app from where radon level data as well as radon fluctuation figure can be obtained should be installed. The “RadonEye” app, which icon is illustrated in Figure 16, can be installed both on Android and iOS. In this experiment, the app was installed on iPad Air 3, the device which supports iOS.



Figure 16 The icon of the "RadonEye" app in App Store.

All units of RadonEye should be cleared from previous recorded data prior starting the experiment. Prior the real experiment, all six units of RadonEye were tested. All units were plugged and started at 18:43 CEST on 23rd of May. All units were connected to iPad, and data from all devices was cleaned. The first data appeared at 19:43 CEST since the time interval was set at one hour.

The data from RadonEye units was retrieved after 24 hours from the time all units were plugged, i.e at 18:43 CEST on 24th of May. The graph which can be obtained from the app looks in the way shown in Figure 17, where y-axis depicts radon concentration whereas x-axis is recorded time in hours:

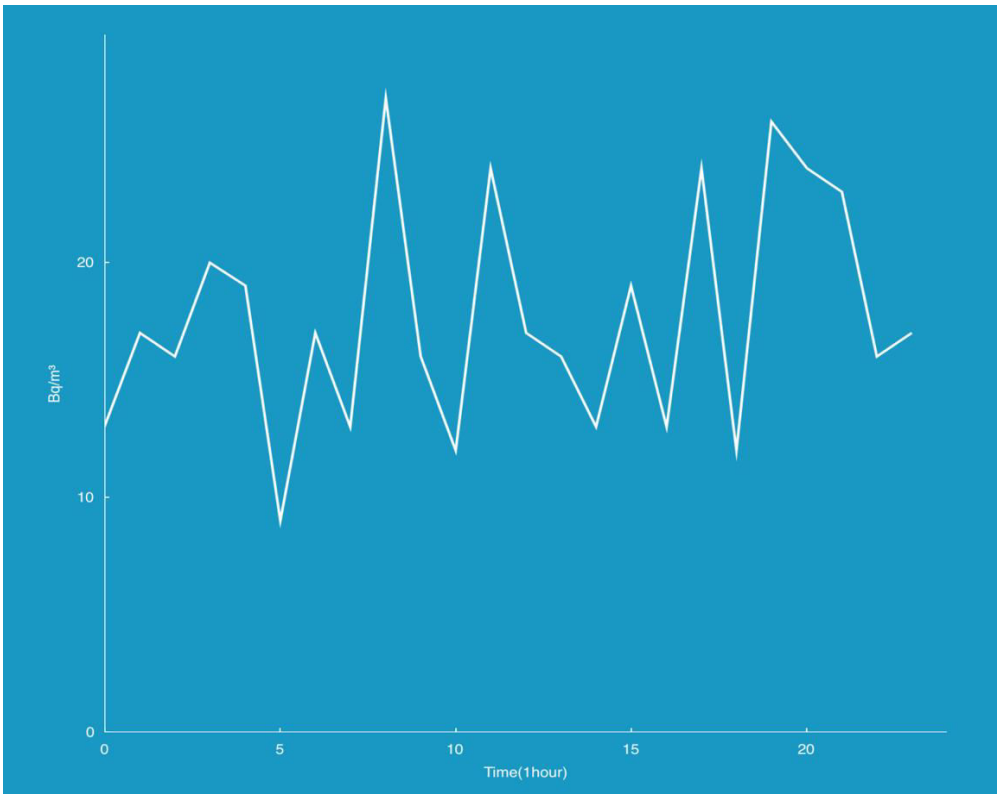


Figure 17 The graph which can be observed in "RadonEye" app for RE21910250130. Graph was retrieved at 18:43 CEST.

Nevertheless, despite the fact that the figure for radon concentration can be seen in the app, it is also possible to download the data in the form of the text, where the first number denotes the radon concentration which was recorded after one hour since RadonEye was plugged in. The graph below, i.e. Figure 18, was done manually based on the hourly radon concentration data obtained from all six units of RadonEye.

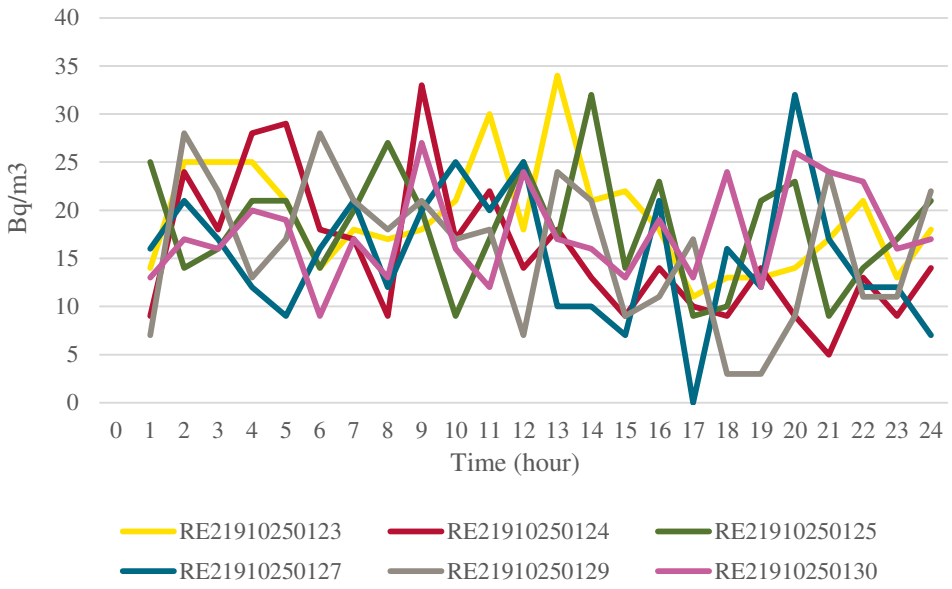


Figure 18 Radon level figures for all six units of RadonEye during 24 hours of measurement

It is important to note that RE21910250127 at the 17th hour did not record any radon level and showed 0 Bq/m^3, and zero level is less likely to reflect the real level of radon since there is a background radiation in the dwellings, as well as outside. In addition, as it can be seen, other devices recorded certain levels of radon, i.e. zero level is an error of the software rather that a real radon level. Thus, the reason of missing data can be due to the errors in the software embedded in RadonEye. Therefore, in order to avoid a missing data, two units of RadonEye should be placed in the same location in order to minimize the probability of the missing data. Therefore, one set should comprise of two units of RadonEye and one unit of Rapidos, and the set itself is shown in Figure 19. However, for one of the locations, namely storage room at Linnaeus University, two units of RadonEye and two units of Rapidos were used.



Figure 19 Representation of the set comprised of two units of RadonEye and one unit of Rapidos.

Three sets were placed in three different locations. After testing all six equipment and confirming that all units work properly, except the fact that there might be missing data for RadonEye which will be compensated by placing two units in the same set, the real experiment can be started. It is important to note that all data should be cleared again.

The radon level measurement for all three locations started on 25th of May. First location was a first-floor apartment at Georg Lückligs väg 10. Here, two units of RadonEye with RE21910250123 and RE21910250124 reference numbers respectively, and one unit of Rapidos with 292123-7 reference number, were placed. The starting time for two units of RadonEye was 12:57 CEST, and the protective cover of Rapidos was removed at 13:01. Due to the requirements for placement position for Rapidos, i.e. at least 25 cm distance from floor, ceiling, or wall, all three units were placed at the height of 87 cm. All three units were placed in the corner of the room at 48 cm from one interior wall, and 43 cm from another interior wall. According to another requirement for the placement of Rapidos, i.e. 1.5 m away from ventilation, window, and exterior door, all three devices were placed at sufficient distance from the aforementioned items.

The second location was a storage room in the basement of M-building at Linnaeus University. Here, two units of RadonEye with RE21910250125 and RE21910250127 reference numbers respectively, and two units of Rapidos with 378754-6 and 413399-7 reference numbers respectively, were placed. The starting time for two units of RadonEye was 14:10 CEST, and the protective covers of both units of Rapidos were removed at 14:11. All four units were placed at the height of 71 cm, and 43 cm away from interior wall. Ventilation and door were at substantial distance from devices, and there is no window in the storage room.

The third location was a first-floor apartment at Georg Lückligs väg 18. Here, two units of RadonEye with RE21910250129 and RE21910250130 reference numbers respectively, and one unit of Rapidos with 803652-7 reference number, were placed.

The starting time for two units of RadonEye was 14:41 CEST, and the protective cover of Rapidos was removed at 14:43 CEST. All three devices were placed at the height of 45 cm, and 55 cm from interior walls. The distance between devices and ventilation, door and windows is sufficient.

For short-term radon data measurement, at least seven days should pass since the data measurement is started. The experiment finished on the 3rd of June for all locations, i.e. the radon measurement took place for 9 days.

For the first location, i.e. apartment at Georg Lückligs väg 10, the data measurement finished at 15:34 CEST: data from RadonEye units was derived, and Rapidos was put to the paper bag which was given by the Swedish Radiation Safety Authority. The following graph, i.e. Figure 20, was obtained for the first location:

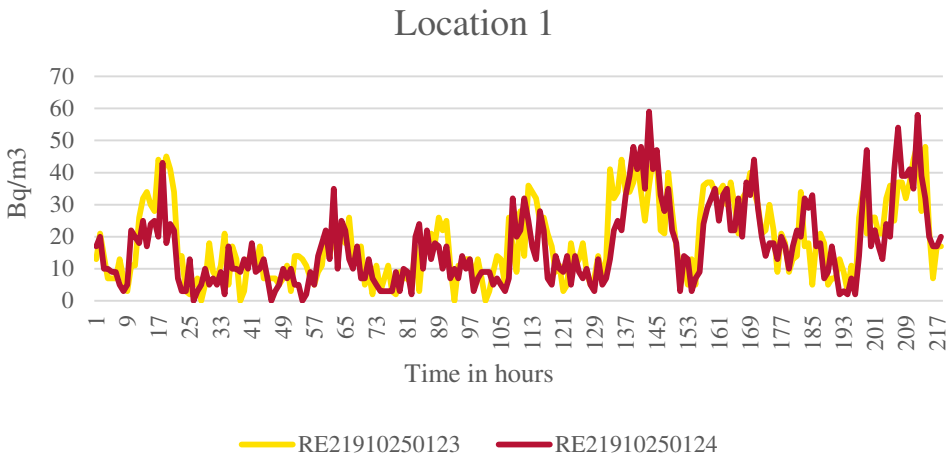


Figure 20 The graph obtained from RadonEye devices in the first location after 9 days of radon measurement.

For the second location, i.e. storage room, the radon measurement was finished at 15:10 CEST. This location is of particular interest since radon level was significantly high first three nights in a row, and then the figure dropped. As it can be seen from Figure 21, the highest concentration registered by RE21910250125 was 513 Bq/m^3 whereas for RE21910250127 it was 526 Bq/m^3, and both values exceed the reference level. Despite the fact that after three nights the levels dropped significantly, there was still a tendency of elevated radon levels at night. The probable reason behind is that the ventilation was reduced at nights.

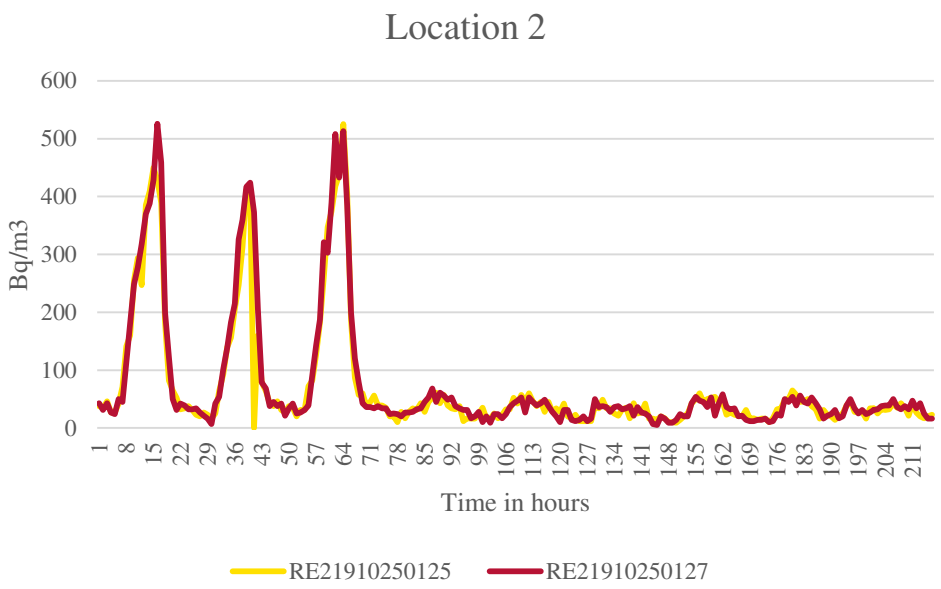


Figure 21 The graph obtained from RadonEye devices in the second location after 9 days of radon measurement.

Another interesting observation is that at 41st hour, one of the devices, namely RE21910250125, recorded 0 Bq/m^3 whereas another device, namely RE21910250127, recorded 373 Bq/m^3. It once more reaffirms the fact that several RadonEye devices should be put at the same location in order to avoid missing data. For the third location, namely Georg Lückligs väg 18, the radon measurement finished at 16:46 CEST.

For the third location, namely Georg Lückligs väg 18, the radon measurement finished at 16:46 CEST. The obtained data is illustrated in Figure 22.

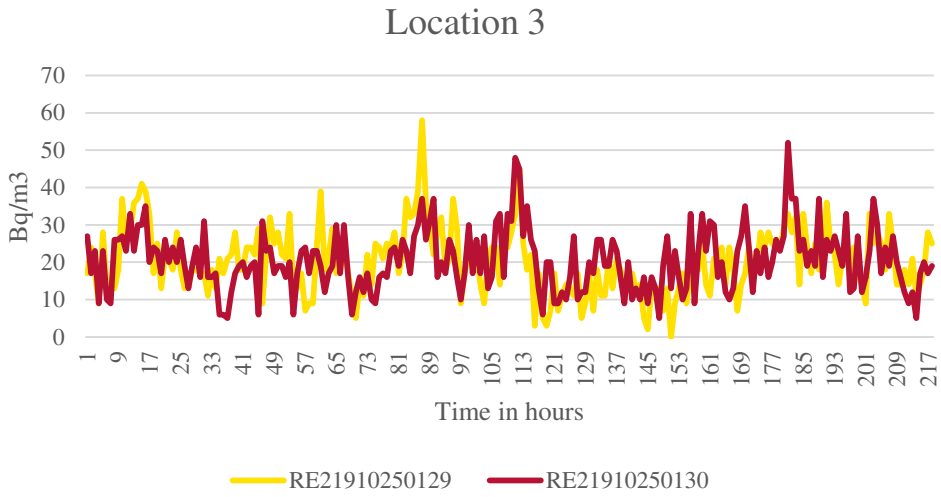


Figure 22 The graph obtained from RadonEye devices in the third location after 9 days of radon measurement.

All devices, i.e. six units of RadonEye and 4 units of Rapidos were sent to the Swedish Radiation Safety Authority for further comparison and analysis.

4 Radon management and safety

The Swedish Radiation Safety Authority is the national body that among other things sets requirements for activities and practices involving radiation. In terms of naturally occurring radioactive materials and radon in workplaces, the authority has issued two special regulations, SSMFS 2018:4 and SSMFS 2018:10. The following is a brief summary of these regulations, the full version of which is available on the SSM website [141].

4.1 SSMFS 2018:10 Regulations on radon in workplaces

These regulations [142], contain provisions to be observed by the activity operating in an environment with ionizing radiation, the person who employs someone in such an activity, or self-employed person, where:

1. The radon content in air exceeds the reference level provided in Chapter 3, Section 6 of the Radiation Protection Ordinance (2018:506). The reference level for radon provided in the aforementioned document is 200 Bq/m^3, which is an annual average activity concentration, in dwellings, places accessible to the public and at work;
2. the annual average radon concentration in air cannot be identified but is expected to exceed the reference level more than temporarily; or
3. exposure to radon in total from several temporary workstations or workplaces is expected to be equivalent to exposure to a radon content in air exceeding the reference level during working hours.

Apart from aforementioned points, employees who are at risk of having an annual radon exposure exceeding 0.72 MBq h/m^3 during working hours should be identified. Workers under the age of 18 also shall not be at risk of receiving an annual radon exposure exceeding aforementioned number during working hours.

In the case if radon content exceeds above-mentioned reference level, the Swedish Radiation Safety Authority should be notified, unless there are measures which are expected to be taken which will reduce the radon content to below the reference level. The following scheme illustrates a simplified general approach to radon regulations in the workplace:

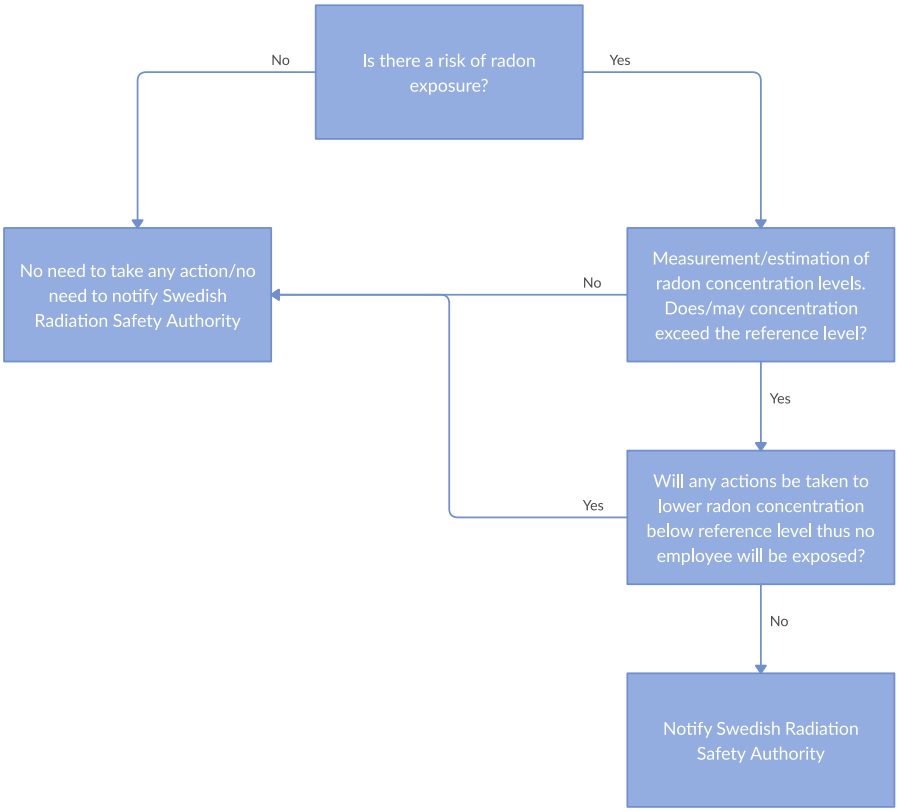


Figure 23 Simplified general approach to radon regulations in the workplace.

The notification shall be made on a form provided by the Authority and shall contain all required information. Activities that are covered by the provisions of these regulations within the Swedish Armed Forces, the Swedish Defense Materiel Administration, the Swedish Fortifications Agency or the National Defence Radio Establishment agencies must be reported to the Defence Inspectorate for Health and Environment. The notification shall contain the information set provided in Annex 1 SSMFS 2018:10 [142]. In the case if there are changes in the activities related to the information which was provided in the notification to the Swedish Radiation Safety Authority, the Authority shall be notified as soon as possible about the changes.

4.2 SSMFS 2018:4 Regulations on naturally occurring radioactive material and building material

The regulations described in [143] contain provisions that shall be applied to those who handle NORM, including building materials. Handling of NORM with an activity concentration exceeding 1 kBq/kg of dry matter per radionuclide in the uranium or thorium series, and which is not exempted from the permit requirement according to §6 of [143], shall be notified to the Swedish Radiation Safety Authority. Due to the fact that radon is included in aforementioned series, and due to the fact that handling of NORM from oil and gas field is not included in §6, namely it is not:

1. the handing of alum shale-based lightweight concrete,
2. the handing of soil or rock masses,
3. holding, storing or in any way non-physically handling of burned alum shale or phosphogypsum,
4. the handing of unprocessed mineral samples included in a geological collection,
5. the handing of ceramic utensils that are intended for household or ceramic ornaments or,
6. the handing of water filters in a single household,

handling of oil and gas NORM exceeding the exemption level shall be notified to Radiation Authority. A notification in accordance shall be made on a form provided by the Swedish Radiation Safety Authority and shall contain the information stated in this. The notification must be renewed within five years from the date the last notification was made if the handling is still ongoing and nothing else has been notified by the Radiation Safety Authority. Changes in the activities relating to the information in the notification must be notified to the Radiation Safety Authority as soon as possible.

In the case if activity concentration of oil and gas NORM is below 1 kBq/kg of dry matter per radionuclide, it can be released. Nevertheless, in the case if activity is above the reference level, it is prohibited to dilute naturally occurring radioactive material if the purpose of this is to enable the material to meet clearance level. The Radiation Safety Authority may authorize the mixing of NORM with non-radioactive material for reuse or recycling or for landfill or other disposal if it can be done without posing an unacceptable risk to humans or the environment from the harmful effects of radiation. Nevertheless, in the case if oil and gas NORM activity concentration is more than 1 kBq/kg of dry matter per radionuclide but not more than 10 kBq/kg of dry matter per radionuclide, radioactive material may be placed in a landfill without taking into account the radioactive properties of the material.

The Swedish Radiation Safety Authority may grant an exemption from these regulations if there are special reasons and if it can be done without it being assumed that there is an unacceptable risk of exposure to humans or the environment from harmful effects of radiation.

4.3 Activities towards the reduction of radon level

When it comes to the reduction of radon level in engine room or areas with LNG equipment, several techniques are suggested in this study. In the case if elevated levels of radon have been detected, the ventilation system in the engine room can be upgraded. In addition, the whole system can be checked for the possible leakage, and fixed in the case of LNG escape. In the case if elevated levels of radon are detected in LNG installation areas, then the possible leakage from flanges and coupling should be checked. In the case of detected leakage, the connection system between LNG source and LNG tank on the ship should be changed or fixed.

For NORM decontamination or proper storage of equipment, one can refer to the "Safe and responsible management of spent nuclear fuel and radioactive waste in Sweden" (2015:32) by the Swedish Radiation Safety Authority [144].

5 Results and discussion

In this study, the question of NORM presence in LNG equipment and LNG area was raised and investigated. The matter on possible NORM presence was raised due to the reason that radon is present in the sedimentary rock, and goes up together with natural gas during production process. To our knowledge, no measurement has been carried on the NORM detection in LNG; therefore, the question of NORM presence in LNG equipment and fuel system is open. Further investigation showed that radon and radon progeny might be present in LNG equipment since:

1. The data on radon and radon progeny in LNG was recorded by [11].
2. Several papers are dedicated to the radon and radon progeny concentration measurement in LNG refinery. One is theoretical [8] proposing the overall picture of possible radon and radon progeny accumulation and movement in LNG refinery, showing that NORM presence in LNG refinery may exceed the reference level and shall be therefore investigated. In another work [9] real data on radon progeny concentration was gathered from Dehydration Unit confirming the fact of NORM movement through multiple units and accumulation.
3. The depository nature of radon and radon decay products on the inner surface of equipment. Several papers such as [60], [62], [63] describe the presence of accumulated NORM in oil and gas industry in different forms such as invisible or gray films on the inner part of equipment, as well as in the form of sludge or scale deposition.

Therefore, in this study, for better understanding of the overall picture, all steps starting from the production to the final LNG product including transportation were analyzed. Main treatment processes taking place in LNG refinery, and what happens to radon and radon progeny on each step, were also reviewed. The following assumptions regarding radon and radon progeny concentration in the final LNG product based on the nature of NORM and treatment processes were made:

1. Radon concentration in final LNG product depends on the radon concentration in the initial feed.
2. Radon concentration in final LNG product depends on the natural gas flow in LNG plant.
3. Radon concentration in final LNG product depends on the materials and filters used in treatment processes.

Assumptions regarding radon and radon progeny deposition in the final destination, e.g. LNG tank or LNG terminal, were also made:

1. Sea motion partially prevents from the deposition of radon and radon decay products in LNG tank on ships.
2. Truck motion also prevents from the deposition of NORM but to a lesser extent due to the fact that motions experienced by LNG during transportation by road have less amplitude.
3. For land-based LNG tanks, it is assumed that NORM tends to accumulate more due to the absence of motion, i.e. particles are settled more easily

In addition, different possible locations to be further investigated where radon and radon progeny are more likely to be detected were proposed:

1. Pipeline, particularly elbows, sudden expansion and contraction, valves, and other parts of pipeline where the direction of LNG changes.
2. Pumps, since the data on NORM contamination was recorded for this type of equipment.
3. LNG land-based tank, since, as it has been mentioned above, absence of motion facilitates the deposition process.
4. Other pieces of equipment such as BOG recondenser, AAV, and etc.

Then, several different techniques for radon and radon decay products detection were suggested:

1. The conversion of LNG to gaseous state following certain standard such as ISO 8943:2007, and then the measurement of radon concentration with the help of different techniques such as alpha scintillation cells or electret ion chamber.
2. Wipe metal sampling, or scale deposition sampling following certain standards such as ISO 7503, or following certain procedures proposed by other organizations and institutions.
3. The use of special device capable of identifying radioactive element at once.

Afterward, real radon measurement for the purpose of equipment comparison for further investigation by the Swedish Radiation Safety Authority was carried. Several important points were drawn:

1. Due to the drawbacks in the software, namely the missing data, it is suggested to place several RadonEye devices in the same location in order to reduce the possibility of missing data.
2. There is an inverse relationship between ventilation and radon levels. In other words, better and stronger is ventilation, lower levels of radon are present, and vice versa. As it can be seen from Figure 21, radon concentration levels increased during night times due to the fact that ventilation was reduced.

Finally, in the case if elevated levels of radiation are detected in the engine room or in the territory of LNG facilities, following methods to reduce radon concentration levels are suggested:

1. For elevated levels in the engine room, to upgrade ventilation system and check LNG equipment for the possible leakage, and to fix in the case of leakage.
2. For elevated levels in the territory of LNG installations, to check the possible leakage caused by flanges and other means of connection between LNG source and LNG tank on the ship. In the case of leakage, to fix the system.
3. For NORM decontamination, one can refer to "Safe and responsible management of spent nuclear fuel and radioactive waste in Sweden" (2015:32) national plan report by the Swedish Radiation Safety Authority.

Nevertheless, there are some points regarding NORM presence in LNG systems, engine room and in the territory of LNG installations which are worth discussing:

1. It is assumed that there is a very low probability of high radon and radon decay products concentration presence in engine room. The reason behind is inability of alpha and beta particles to escape through equipment. The possible way of escaping is LNG leakage; however, the leakage may not be detected only in the case if detectors installed in the engine room are not working properly. Since it is assumed that the system is regularly checked, the possibility to detect concentrations exceeding reference level is very low, i.e. there is no hazard associated with radiation in engine room.
2. The same logic is applied to the territory with LNG installations, i.e. only in the case of leakage, for instance from equipment connecting LNG tank on ship to LNG source, elevated radon levels can be detected. Otherwise, there is no hazard to employees engaged into activities in the territory of LNG installations.
3. There is a high probability of hazard associated with maintenance of LNG equipment. The are several reasons behind such as accumulation of NORM on the inner surface of equipment leading to increase in radioactivity level, and the opening of equipment for cleaning and other required procedures where alpha and beta particles can be inhaled thus can pose a threat to the health.

6 Conclusion

In order to detect NORM contamination, several methods are proposed in this study. Certain detectors, such as NaI, can be used for preliminary radiation detection, and techniques for exact NORM identification, such as wipe or scale sampling, as well as sludge, can be used. Besides, LNG can be converted to the gaseous state, and further analyzed for radon presence.

There are several locations where radon and radon progeny concentrations are expected to be the highest. The highest possible accumulation of NORM is expected to be in LNG tank due to the long hold up. Apart from LNG tank, NORM is

expected to be deposited on the inner parts of pipes, particularly on outer edge of elbows, valves, t-joints, sudden contraction and expansion. Shortly, in any region where the direction of LNG flow changes, there is a higher possibility to detect NORM. In addition, pump also can be checked for NORM deposition.

There are numerous techniques that can be used for radon level measurement. However, most of them are for radon measurement in air. In the case of LNG, LNG can be converted to gaseous state, as it has been already mentioned, and then radon levels can be measured via alpha scintillation cells, or via electret ion chamber. Activated charcoal detector can also be used; however, due to numerous factors such as dependence on the humidity, temperature, and radon fluctuations, it is suggested to use alpha scintillation cells or electret ion chamber.

In order to obtain reliable results, it is essential to adhere to certain standards. For instance, in the case if radon measurement will be carried on LNG in the gaseous state, it is suggested to follow ISO 8943:2007 standard where the steps for right sampling of LNG are described. In the case if it is decided to go with wipe sampling for metals technique, it is suggested to follow ISO 7503 standards, or use instructions provided by different institutions and organizations such as [123]. In the case if scale deposition or sludge should be analyzed, the steps provided by the Swedish Radiation Safety Authority should be taken.

Due to the fact that the diameter of radon and radon progeny particles varies a lot, different forces have effect on the particles of different sizes. For very small particles, thermophoresis, Brownian and diffusional forces are the main forces acting on the particles. For larger particles, or for relatively small particles but with high density, gravitational force is the main acting force. Therefore, it is assumed that particles with different size and density behave differently in LNG. Nevertheless, common assumptions regarding NORM deposition on the inner surface of LNG equipment based on the behavior of radon and radon decay products particles can be made, and locations were described in detail in this study.

The matter of radiation is of high importance, and since, to our knowledge, there is no work carried on the NORM measurement in LNG systems, the question is still open. It is essential to ensure a safe work environment, especially for the maintenance workers since maintenance workers are under the highest risk of radiation exposure. This study provides a ground and base for further investigation and measurement of radon and radon decay products in LNG systems and areas of LNG installations.

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